Journal of Human Evolution 149 (2020) 102882

Contents lists available at ScienceDirect

Journal of Human Evolution

journal homepage: www.elsevier.com/locate/jhevol

Meat eating by nonhuman primates: A review and synthesis

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ARTICLE INFO

Article history:
Received 1 January 2020
Accepted 21 August 2020
Available online 24 October 2020

Keywords:
Primates
Meat eating
Nutrition
Chimpanzees
Capuchins
Baboons

ABSTRACT

Most nonhuman primates prey on vertebrates. Meat-eating, defined as ingestion of vertebrate tissue, occurs in 12 families, ≥39 genera, and ≥89 species. It is most common in capuchins (*Cebus* and *Sapajus* spp.), baboons (*Papio* spp.), bonobos (*Pan paniscus*), and chimpanzees (*Pan troglodytes*) and modestly common in blue monkeys (*Cercopithecus mitis*), callitrichids (*Callithrix* spp. and *Saguinus* spp.), and squirrel monkeys (*Saimiri* spp.). It is uncommon in other cercopithecines, rare in other haplorhines and in lemurs, and virtually absent in colobines. Birds are the prey class eaten by the most species (≥53), followed by reptiles (≥48), amphibians (≥38), mammals (≥35), and fish (≥7). Major hypotheses for the importance of meat eating are that it is (1) mainly an energy source, especially (1a) when plant-source foods (PSFs) with high energy return rates are scarce (energy shortfall hypothesis); (2) mainly a protein source; and (3) mainly a source of micronutrients scarce in PSFs. Meat eating bouts sometimes provide substantial energy and protein, and some chimpanzees gain substantial protein from meat monthly or annually. However, meat typically accounts for only small proportions of feeding time and of total energy and protein intake, and quantitative data are inconsistent with the energy shortfall hypothesis. PSFs and/or invertebrates are presumably the main protein sources, even for chimpanzees. Support is strongest for the micronutrient hypothesis. Most chimpanzees eat far less meat than recorded for hunter-gatherers, but the highest chimpanzee estimates approach the lowest for African hunter-gatherers. In fundamental contrast to the human predatory pattern, other primates only eat vertebrates much smaller than they are, tool-assisted predation is rare except in some capuchins and chimpanzees, and tool use in carcass processing is virtually absent. However, harvesting of small prey deserves more attention with reference to the archaeological and ethnographic record.

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1. Introduction

Nearly 40 years ago, Harding (1981) argued that most nonhuman primates are omnivores: their diets come primarily from 'plant-source foods' (PSFs; cf. Milton, 2003a, b), but they also eat animal-source foods (ASFs; cf. Milton, 2003a, b). ASFs are better sources of essential amino acids, and animal protein is more bioaccessible than plant protein (Carpenter, 1994; Layman et al., 2015); meeting protein needs would require less ASF, even though PSFs, especially leaves, can be high in protein (Milton, 2003b). ASFs also are valuable micronutrient sources (Milton, 2003a; Phillips et al., 2015; Payne et al., 2016). Many factors—notably body size variation (Kay, 1984)—constrain reliance on ASFs, and their dietary contributions vary greatly (Harding, 1981; McGrew, 2014). The ASF component of most species' diets consists mostly or entirely of invertebrates, especially arthropods (McGrew, 2014; Rothman et al., 2014), but many eat vertebrates. In a valuable review, Butynski (1982a) reported vertebrate-eating in 9 nonhuman primate families (and in Tupaidae), 12 subfamilies, 29 genera, and 36 species. However, vertebrates usually contributed little to overall food intake, and chimpanzees (*Pan troglodytes*) and baboons (*Papio* spp.) accounted for 80.7% (363/450) of meat-eating cases. No comprehensive review of meat eating has appeared since, despite the greatly expanded literature on use of animal carcasses by early hominins, the debates about the roles of hunting, scavenging, and meat sharing in human social and life history evolution (e.g., Wrangham et al., 1999; Kaplan et al., 2000; Bunn, 2007; Hawkes and Coxsworth, 2013; Zink and Lieberman, 2016) and the relevance of chimpanzees to these issues (e.g., Moore, 1996; Pickering and Dominguez-Rodrigo, 2010; Sayers et al., 2012; Stanford, 2012; Wood and Gilby, 2017; Thompson et al., 2019).

Here I provide an updated review of meat eating in nonhuman primates. Following Teleki (1975), I define meat as vertebrate tissue

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https://doi.org/10.1016/j.jhevol.2020.102882
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(muscle, viscera, brains, etc.) and eggs from birds and other vertebrates. Egg eating does not involve active pursuit, although it can entail risk of injury by nest defenders (Robinson, 1997). But predation need not involve active pursuit; examples include opportunistic seizure of cached Thomsen's gazelle (*Gazella thomsoni*) fawns by baboons (e.g., Harding, 1975), of red duiker (*Cephalophus natalensis*) fawns by chimpanzees (e.g., Mitani and Watts, 1999), and of coati (*Nasua narica*) pups by white-faced capuchins (*Cebus capucinus*; e.g., Rose, 1997). I particularly focus on baboons, capuchins (*Cebus* spp. and *Sapajus* spp.), and chimpanzees, in which predation on vertebrates is most frequent and best described. I use the data to assess 3 hypotheses about the nutritional importance of meat eating and briefly consider their implications for debates about the role vertebrate tissue consumption in human evolution. These hypotheses, which could apply differently to different cases, are:

1) Individuals eat meat because it is a valuable energy source (Hamilton and Busse, 1978; Boesch, 1994a). If its value is solely energetic, it should provide higher net energy intake than alternative foods. The specific 'energy shortfall hypothesis' proposes that meat is particularly important when other sources of easily assimilated energy (notably fruit) are scarce and predicts that hunting frequency and meat intake should vary inversely with the availability of such foods.
2) Nonhuman primates eat meat to obtain protein.
3) Meat is primarily a source of micronutrients, especially those scarce in many PSFs (Hamilton and Busse, 1978; Milton, 2003a, b; Gilby et al., 2008; Tennie et al., 2009, 2014).

McGrew (2014) and Rothman et al. (2014) recently reviewed primate insectivory. I do not review scavenging because (a) it is rare (Watts, 2008; Hosaka, 2015a); (b) Nakamura et al. (2019) recently described confrontational scavenging of a leopard kill by chimpanzees and evaluated its relevance to debates about early hominin behavior and (c) Pobiner (2020) reviewed these debates and reiterated that increased nutrient harvesting via scavenging by Plio-Pleistocene hominins crucially departed from ASF use by other primates (cf. Thompson et al., 2019). I have also not included cannibalism, although it occurs in multiple species and might mainly be nutritional (Hiraiwa-Hasegawa, 1992; Hosaka, 2015a; Fedurek et al., 2020); it also deserves a separate review. I briefly consider meat transfers below, but a complete review of the relevant literature is not possible here; the same holds for the literature on cooperative hunting.

2. Materials and methods

Except for unpublished personal data (e.g., observation of predation by olive baboons (*Papio anubis*) on red duiker, I used data from general reviews by Harding (1981) and Butynski (1982a), reviews of particular taxa (e.g., Harding, 1975; Rose et al., 2003; Fichtel, 2012), and primary sources. I located sources via published reference lists, searches of Web of Science and PubMed using terms like 'primates' and 'meat eating', and content searches of *Pan Africa News* and *Neotropical Primates*. I omitted some references on chimpanzees that did not add to data on prey repertoires, hunting rates, or meat intake. Russon et al. (2014) reviewed data on predation on aquatic fauna, including fish. I included fish eating when researchers saw capture of live fish or found fish bones in feces from animals that could not have obtained fish from humans.

Few nutritional and energy data are available for prey of chimpanzees (Tennie et al., 2014) or other species. I used data on wolf prey (Bosch et al., 2015), white-tailed deer (Pennsylvania State Game Commission, 2020), and several African bovids (Ntiamoa-Baidu, 1997; Hoffman and Ferreira, 2004; Hoffman, 2008; Kyriacou et al., 2016; FAO, 2018) to estimate gross energy and protein intake by chimpanzees, and I estimated the proportional contribution of meat to total daily energy expenditure (TEE) using data in Pontzer et al. (2014) and Uwimbabazi et al. (2018; Supplementary Online Material [SOM]; SOM S1 and Table S1). I used data on gross energy and protein values for leaves, fruit, and figs from 3 chimpanzee research sites (Wrangham et al., 1991; Conklin and Wrangham, 1994; Potts et al. 2015; K.B. Potts, unpubl. data; Uwimbabazi et al., 2018; Bryson-Morrison et al., 2020) to estimate PSF equivalents to the energy and protein provided by meat at several chimpanzee research sites (SOM S1 and Table S2).

3. Results

3.1. Taxonomic distribution of vertebrate predation

Appendix 1 lists all primates for which I found references to meat-eating along with prey taxa. The list now includes 12 families, with Atelidae, Colobidae, and Pitheciidae added (with just a few observations each), 38 genera, and approximately 89 species (the number depends on taxonomic choices—e.g., does *Papio* contain multiple species or one species with multiple subspecies?). It is presumably incomplete because of undersampling. For example, all callitrichids other than pygmy marmosets (*Cebuella pygmaea*) for which I could find data eat small lizards and/or frogs and many eat bird nestlings and/or eggs, but we lack data for many species.

Among strepsirrhines, meat eating is most widespread in Lorisidae and Galagidae. It also occurs in several dwarf and mouse lemurs. Among other Lemuridae, it is known only in *Eulemur fulvus*, and it is seemingly absent in Indriidae.

Tarsiers appear distinct in being completely faunivorous. However, only *Tarsius bancanus* definitely eats vertebrates (Niemitz, 1973, 1979, 1984; Gursky, 2007). Vertebrate prey includes snakes, lizards, birds, and several species of Pteropodid bats. Captive individuals readily eat fish, suggesting that predation on fish occurs in the wild (Niemitz, 1979), but no published confirmation is available.

Among Platyrrhines, meat eating is widespread in marmosets and tamarins, capuchins, and squirrel monkeys (*Saimiri* spp.). Callitrichids and squirrel monkeys mostly eat amphibians, reptiles, and birds, but black-penciled marmosets (*Callithrix pencillata*) prey on agile opossums (*Gracilianus agilis*; de Camargo et al., 2017) and Central American squirrel monkeys (*Saimiri oerstedii*) prey on tent-making bats (*Artibeus watsoni*; Boinski and Timm, 1985). Capuchins eat amphibians, reptiles, birds, and mammals. Meat eating is uncommon in pitheciids and rare in Atelidae, for which only a few reports of predation on frogs, small rodents, and snakes by woolly monkeys (*Lagothrix lagotricha* and *Lagothrix flavicauda*; Di Fiore, 2004; Fack et al., 2018) and on domestic chicken adults and bird eggs and nestlings by howler monkeys (*Alouatta* spp.; de Thoisy and Parc, 1999; Lawrence, 2003; Bicca-Marques et al., 2009, 2014) exist.

Baboons stand out among cercopithecoids for the breadth of their prey repertoires and because all taxa for which data exist (olive baboons; yellow baboons, *Papio cynocephalus*; Hamadryas baboons, *Papio hamadryas*; Guinea baboons, *Papio papio*; chacma baboons, *Papio ursinus*) eat other mammals. Most other papionins eat meat rarely, if at all. Two reviews of variation in macaque (*Macaca* spp.) diets (Richter et al., 2013; $n = 14$ species; Cui et al., 2019; $n = 9$) reported that ASFs accounted for 2–3% to nearly 46% of feeding time, but little or no time was devoted to vertebrate prey. For example, Sulawesi crested macaques (*Macaca nigra*) spent 26–37.3% of monthly feeding time eating ASFs but ate adult vertebrates only 8 times and bird eggs only twice during O'Brien and

Kinnaird's (1997) study, and Japanese macaques commonly eat invertebrates, but rarely eat vertebrates (Suzuki et al., 1991: 421).

ASFs are generally unimportant in colobine diets, although a few exceptions exist (Rothman et al., 2014). For example, Wachter et al. (1997) saw no feeding on ASFs by West African red colobus (*Procolobus badius*) at Taï (Ivory Coast) in 2748 observation hours and in a review of data on 22 colobine species at 45 temperate, tropical, and alpine habitat sites, Tsuji et al. (2013) found reports of ASF use by only 4 species, at 5 sites, for 1.5–3.6% of feeding time. Meat eating is known only from a few cases of feeding on bird eggs and flying squirrels by *Rhinopithecus bieti* (Grueter et al., 2009; Appendix 1). By analogy with ruminants, colobines can probably ferment ASFs in their forestomachs, but they can satisfy protein needs through digestion and microbial fermentation of PSFs (Kay and Davies, 1994).

Vertebrate predation by hylobatids and orangutans is uncommon. No unequivocal accounts exist for gorillas. Hofreiter et al. (2010) found monkey and duiker DNA in western gorilla (*Gorilla gorilla*) feces at Loango (Gabon); they noted that meat eating might have led to its presence although no behavioral evidence of vertebrate predation by gorillas exists but thought exogenous contamination or endogenous contamination via ingestion of ants that had fed on carcasses were more likely explanations. The only meat eating by bonobos (*Pan paniscus*) known to Butynski (1982a) was Badrian and Badrians (1977) account of predation on duiker; the known range of prey now includes birds and many mammals (Appendix 1). Chimpanzees show by far the widest range of prey of all nonhuman primates (Appendix 1; see also below).

3.2. Taxonomic distribution of prey

Birds are the most widespread class of prey, with eggs, nestlings, and/or adults included in diets of 53 species in 27 genera. Predation on bird nests is common in some callitrichids (e.g., common marmosets, *Callithrix jacchus*, are nest predators on at least 10 species). It is especially common in capuchins, although this may reflect sampling bias, given that many accounts are from studies of nest predation (e.g., Robinson, 1997 for Cocha Cashu, Manu National Park, Peru; Menezes and Marini, 2017): as genera, *Cebus* preys on at least 21 species (11 by white-faced capuchins) and *Sapajus* on at least 23 (21 by black-capped capuchins, *Sapajus apella*). Forty-eight species in 24 genera eat reptiles (lizards, tortoises, turtle eggs, snakes); 38 in 18 genera eat amphibians (frogs); and 35 in 18 genera eat mammals. Mammalian prey include rodents, macroscelids, Eulipotyphla, pholidota, artiodactyls, chiropterans, lagomorphs, carnivores, and primates. Among New World monkeys, capuchins eat the widest range of mammals: *Cebus* species prey on at least nine species (including at least five on which white-faced capuchins prey) and *Sapajus* species prey on at least 11 (including at least 10 on which black-capped capuchins, *S. apella*, prey). Baboons as a genus prey on at least 21 wild mammal species from 5 orders, by far the widest repertoire among cercopithecoids; olive and yellow baboons prey on 9 species each. Olive and chacma baboons also sometimes prey on immature domestic sheep and goats (Butler, 2000; Sogbohossou et al., 2011), and yellow baboons are also known to prey on domestic goats (Holmern et al., 2007). Mandrills (*Mandrillus sphinx*), lion-tailed macaques (*Macaca silenus*), Sulawesi crested macaques, and one population of pig-tailed macaques (*Macaca nemestrina*) are the only other papionins known to eat mammals. However, frequent predation on rats by pig-tailed macaques foraging in Malaysian oil palm plantations (a recent innovation: Ruppert et al., 2018; Holzner et al., 2019) highlights the ecological flexibility of this and many other *Macaca* species. Many guenons (*Cercopithecus* spp.) occasionally eat meat, as do *Chlorocebus* spp., but only blue monkeys (*Cercopithecus mitis*) and green monkeys (*Chlorocebus sabaeus*) are known to eat mammals. Predation on mammals is unknown in hylobatids and only occasional in orangutans. In contrast, chimpanzees prey on at least 45 species from 6 orders, and bonobos prey on at least 16 from 5 orders, more than all nonhuman primate except chimpanzees (Appendix 1).

Predation on other primates (Appendix 1) includes predation by dwarf lemurs (*Mirza coquereli*) on mouse lemurs (*Microcebus murinus*); by yellow and chacma baboons on vervets (*Chlorocebus pygerythrus*); by yellow baboons on galagos (*Galago senegalensis*); by black-capped capuchins on owl monkeys (*Aotus lemurinus*; Carretero-Pinzon et al., 2003) and on brown (*Callicebus brunneus*; Lawrence, 2003) and dusky titis (*Callicebus molloch*, Lawrence, 2003; Sempaio and Ferrari, 2005); by golden-bellied capuchins (*Sapajus xanthosternos*) on common marmosets (Sousa et al., 2013); by Bornean orangutans (*Pongo abelii*) on slow lorises (*Nycticebus coucang*); by bonobos on 3 cercopithecine species, 1 colobine, and 1 galago; and by chimpanzees on 21 cercopithecoid species and at least 4 of Lorisiformes. Bastos et al. (2018) reported that common marmosets in a Brazilian Atlantic coast forest fragment show typical antipredation behavior (vigilance or flight) in response to blonde capuchin (*Sapajus flavius*) calls, to call playbacks, and to approaching capuchin groups. *S. xanthosternos* might limit common marmoset populations in young growth forests that lack other predators (Hilario and Ferrari, 2015), and heavy predation by chimpanzees at Ngogo (Kibale National Park, Uganda) led to a major decline in the local population of red colobus (*Piliocolobus tephrosceles;* Teelen, 2007, 2008; Watts and Amsler, 2013; Watts and Mitani, 2015; cf. Stanford, 1995 for Gombe).

Primates terrestrial enough to have opportunities to prey on fish are probably rarely in suitable situations (Hamilton and Tilson, 1985; Stewart et al., 2008). Russon et al. (2014) concluded that evidence for fish eating was firm for 9 species, with bonobos a probable tenth. Active capture was restricted to chacma baboons, rehabilitant Bornean orangutans (*Pongo pygmaeus*) who regularly saw humans fishing and long-tailed macaques (*Macaca fascicularis*). Observations of a chimpanzee at Bossou (Guinea) eating a 'small fish' (Sugiyama and Koman, 1987) and of opportunistic fish-eating by Anubis baboons (Matsumoto-Oda and Collins, 2016) should be added (Appendix 1). Zeeve (1993) found fish bones in feces of Allen's swamp monkey (*Allenopithecus nigriviridis*) and De Brazza's monkey (*Cercopithecus neglectus*) and captive *T. bancanus* eat fish provided by humans (Niemitz, 1979).

3.3. Overview of meat-eating frequency and intake

Many references in Appendix 1 simply state that study animals sometimes ate vertebrates, implying that vertebrate prey were minor dietary components. Invertebrates accounted for most prey captures (Fig. 1) and most feeding time (Table 1) devoted to ASFs, and for only small percentages of all feeding records (Table 1), in species for which quantitative data on amphibian and reptile predation are available. However, frogs and lizards were 13.1% of all feeding records for a buffy-headed marmoset (*Callithrix flaviceps*) group (Hilario and Ferrari, 2010), and Goeldi's monkeys (*Callimico goeldii*) spent up to 9% of monthly feeding time eating lizards during Porter's (2001) study. Also, predation on nests of colonially nesting and/or seasonally breeding birds can lead to temporarily elevated intake of eggs and nestlings (e.g., vervets: Skinner and Skinner, 1974; olive capuchins, *Cebus olivaceous*: Robinson, 1986; black capped capuchins: Robinson, 1997). Feeding time may not accurately assess relative intake, and size disparities between vertebrate and invertebrate prey limit the usefulness of percent prey capture data (Tennie et al., 2014).

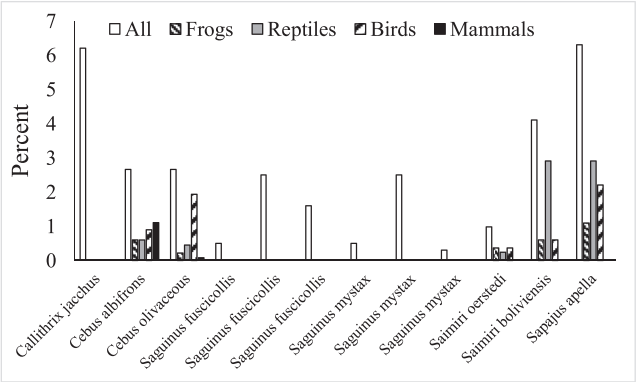

Figure 1. Percentage of all prey captures of vertebrates by 8 species of New World Monkeys. In some cases, researchers reported only total numbers, not breakdowns by taxonomic category. References: *Callithrix jacchus* (Abreu et al., 2016); *Cebus albifrons*, *Saimiri oerstedii*, *S. boliviensis*, *Sapajus apella* (Janson and Boinski, 1992); *Saguinus fuscicollis*, *S. mystax* (Peres, 1993; Heymann et al., 2000; Nadjafzadeh and Heymann, 2008).

The most extensive quantitative data on meat eating are from white-faced capuchins, baboons, and chimpanzees. Capuchins get vertebrate ASFs mostly by raiding bird, squirrel, and coati nests for nestlings, eggs, and pups; they sometimes capture juvenile and adult squirrels, other small rodents, frogs, and lizards (Appendix 1), and some prey on other primates (above). Much intraspecific and interspecific variation exists, and predation frequencies are low except for white-faced capuchins at Santa Rosa and Lomas Barbudal (Costa Rica), particularly during the coati birth season (Rose, 1997, 2001; Rose et al., 2003). At Santa Rosa, hunting and capture rates for all prey and for mammals were within the chimpanzee range (Table 2; Fedigan, 1990; Rose, 1997, 2001; Rose et al., 2003). Also, success rates of 50–81% for squirrel hunts were within the range of success rates for hunts by chimpanzees (Hosaka et al., 2020). Meat accounted for 1–3% of feeding time at Santa Rosa (Rose, 1999) and 1% at Lomas Barbudal (Perry, 2008, Table 1).

Baboon hunting rates and prey repertoires vary considerably. Fawns of Thompson's gazelle and impala (*Aepyceros melampos*) were among the most common prey at Gilgil (Kenya; Harding, 1973, 1975; Strum, 1981) and Amboseli (Kenya; Altmann and Altmann, 1970; Hausfater, 1976), but neither occurs at Moremi (Botswana), where cape hares (*Lepus capensis*) were the main prey (Hamilton, 1973; Hamilton and Busse, 1982). Hamadryas baboons at Filoha (Ethiopia) prey on at least ten species of lizards, birds, and mammals, although infrequently (Swedell, 2006; Schreier et al., 2019). Olive baboons at Mikumi (Tanzania) captured only 4.3 vertebrate prey (bird eggs, lizards, and mammals) per capita annually during Rhine et al's. (1986) study (Table 2); the authors concluded that meat eating had little nutritional importance there or at other sites. But olive baboons at Gilgil, where large mammalian carnivores were mostly absent, captured 15 mammal prey per month during Harding's (1981) study; this exceeds average monthly values reported for chimpanzee study sites (Table 2). Predation rates at Gilgil were similarly high during Strum's (1981) study.

Chimpanzee hunting and prey capture rates vary greatly among sites and within sites over time (Newton Fisher, 2015, Table 2). For example, the number of known hunts per month varied from 0 to 20 over a 25-year period at Ngogo (Kibale, Uganda), and prey captures varied from 0 to 48 per month (Watts and Mitani, 2015, unpubl. data; Table 2). The mean number of months varied from less than one at Sonso (Budongo Forest Reserve, Uganda) between 1994 and 2002 (Hobaiter et al., 2017) to 8 hunts per month at Kasekela (Gombe, Tanzania) in the early 1990s (Stanford et al., 1994a, Table 2). Except for galago hunting at Fongoli (Pruetz and Bertolani, 2007; Pruetz et al., 2015), hunting is relatively uncommon in West African woodland-savanna-farmland mosaics (Pruetz et al., 2015; Caiquene-Cadique, Guinea-Bissau:; Bessa et al., 2015). Observational data on chimpanzees in other savanna–woodland habitats (e.g., Mt. Assirik, Senegal: McGrew et al., 1979, 1988; Semliki, Uganda: Hunt and McGrew, 2002; Issa, Tanzania: Moore et al., 2017) are limited. However, in a comparative review of evidence for vertebrate eating derived from fecal samples, Moore et al. (2017) concluded that savanna chimpanzees may hunt as often as those in forested habitats despite lower primate species diversity and lower chimpanzee densities at savanna sites, but they mostly capture smaller prey and probably eat less meat.

Red colobus (*P. tephrosceles*; *P. badius*) are typically the main chimpanzee prey where they are sympatric. They accounted for 64% of hunts and 83% of captures over 20 years at Ngogo (Watts and Mitani, 2015). Redtail monkeys (*Cercopithecus ascanius*) and gray-cheeked mangabeys (*Lophocebus albigena*) were much more

Table 1
Feeding on vertebrate prey as a percentage of all feeding records or all feeding time.

Species	% Records	Prey class	Reference
Callithrix flaviceps	13.1	All	Hilario and Ferrari (2010)
Callithrix geoffroyi	0.8	All	Passamani and Rylands (2000)
Nomascus concolor	<0.1	All	Fan et al. (2013)
	% Time		
Leontopithecus chrysopygus	0.6[a]	Frogs	Keuroghlian and Passos (2001)
Callimico goeldii	1.0–9.0[b]	Lizards	Porter (2001)
Cebus capucinus	1.0–3.0[b]	All	Rose (1997) [c]
	1.0	All	Perry (2008) [d]
Lagothrix lagotricha	0.13	All	Di Fiore (2004)
Macaca fascicularis	0.35–0.9[b]	Frogs	Son (2003)
Macaca fuscata	<1.0	All	Hanya (2004)
Saguinus fuscicollis	<1.0	Frogs	Porter (2001)
Pan troglodytes	0.1	Mammals	Wrangham (1977)
	<5.0	Mammals	Goodall (1986)
	<3.8	Mammals	Thompson and Wrangham (2008)
	0–12.3[b]	Mammals	Watts and Mitani 2015; Watts unpubl.

ASFs, animal-source foods.
[a] First month of rainy season only, otherwise 0; 12% of feeding time devoted to ASFs.
[b] Monthly range.
[c] Santa Rosa.
[d] Lomas Barbudal.

Table 2
Predation/hunting rates. For chimpanzees, data from Sonso, Kahuzi-Biega, Fongoli, Mahale, and Ngogo include all prey species; data from other sites as noted.

Species	Hunt rate	Capture rate	Site	Reference
Cebus capucinus	3.66/100 h	5.35/100 h all prey	Santa Rosa	Rose (1997)
		2.69/100 h mammals		
		1.98/100 h birds/eggs		
		1.14/100 h squirrels		
		1.54/100 h coati pups		
Sapajus libidinosus	0.3/100 h	0.3/100 h snakes	Piaui (2 sites)	Falotico et al. (2018)
Papio anubis		1/22 h, 15 prey/month, all prey	Gilgil	Harding (1973)
Papio anubis		4.34 items/individual/y	Mikumi	Rhine et al. (1986)
Pan paniscus	2/month		Lui Kotale	Fruth and Hohmann (2018)
	1/month	1/month	Iyema[a]	Wakefield et al. (2019)
Pan troglodytes	0.43/month		Sonso[b,c]	Hobaiter et al. (2017)
	1.04/month		Sonso[b,c]	Hobaiter etal.(2017)
	1/3.8–13.9 days		Kanyawara[d,e]	Gilby and Wranga (2007)
	0.3/100 h (1.1/month)	0.8/month	Kanyawara[d,f]	Gilby et al. (2015)
	2.75/month	4.75/month	Kanyawara[d,e]	Gilby et al. (2017)
	2.5/month	≥2.5/month	Kahuzi-Biega	Basabose and Yamagiwa (1997)
		c. 2.5/month	Fongoli[g]	Pruetz et al. (2015)
	4.5–10/month	2.79–6.21/month	Taï[d]	Boesch and Boesch (1989)
		1.41/month	Kasekela[h,i]	Wrangham and Bergmann-Riss (1990)
		0.79/month	Kahama[i]	Wrangham and Bergmann-Riss (1990)
	8/month	6.48/month	Kasekela[d,j]	Stanford et al. (1994)
	1.51/100 h (5.4/month)	6.4/month	Kasekala[d,k]	Gilby et al. (2015)
	1.56/month		Mahale M Group[a,l]	Takahata et al. (1984)
		5/month	Mahale M Group[m]	Uehara et al. (1992) [p]
	3.88/month	6.67/month	Mahale M Group[n,o]	Hosaka et al. 2001)
	1.91/month	1.44/month	Mahale M Group[p]	Hosaka et al. (2020)
	1.11/100 h (4.0/month)	2.8/month	Mitumba[d,q]	Gilby et al. (2015)
	4.9/month (range 0–2)	7.6/month (range 0–48)	Ngogo	Watts and Mitani (2015), unpubl. data

[a] Successful hunts only.
[b] Estimates for successful hunts corrected by assuming constant rate of unsuccessful hunts.
[c] Sonso estimate of 0.43 hunts/month is for 1994–2002; estimate of 1.04/month is for 2003–2017.
[d] Red colobus only.
[e] Minimum estimate based on length of study.
[f] 1995–2014; Calculated from data in Gilby et al. (2015).
[g] Estimated from data in Pruetz et al. (2015).
[h] 1973–1975.
[i] Data on males only.
[j] 1989.
[k] 1976–2013; calculated from data in Gilby et al. (2015).
[l] 1982–1992.
[m] 1979–1982.
[n] 1981–1990.
[o] Corrected for observation days.
[p] 1965–2010.
[q] 2000–2014; calculated from data in Gilby et al. (2015).

abundant but accounted for only 8.1% and 3.6% of hunts, respectively, and 3.8% and 1.5% of kills. Red colobus are also the main prey at Kasekela and Mitumba (Gombe, Tanzania; Stanford, 1998; Gilby et al., 2006, 2017), Mahale (Tanzania; Hosaka et al., 2001, 2020; Hosaka, 2015b), Taï (Boesch and Boesch, 1989; Boesch and Boesch-Achermann, 2000), and Kanyawara (Kibale; Gilby and Wrangham, 2007; Gilby et al., 2008, 2015, 2017). Sebitoli, a third Kibale site, is the only well-studied site where predation on sympatric red colobus is unknown, even though the local red colobus population density is high (S. Krief, pers. comm.). Guerezas (*Colobus guereza*), the second-most common prey at Ngogo and Kanyawara, are the main prey at Sebitoli and also at Sonso, where red colobus are absent (Newton-Fisher et al., 2002; Hobaiter et al., 2017). Hunting rates seem low in montane forests that also lack red colobus and have low primate species diversity (Kahuzi-Biega: Basabose and Yamagiwa, 1997; Nyungwe: Matthews et al., 2019). However, rates are similarly low at Kanyawara. Hunting was infrequent at Sonso early in research there (Newton-Fisher et al., 2002) but has since become more common as the chimpanzees have increasingly targeted guerezas (Hobaiter et al., 2017). Mahale M group hunted infrequently in the 1960s and 1970s but increasingly more often thereafter as red colobus hunts became more common (Hosaka et al., 2001, 2020; Hosaka, 2015b).

Chimpanzees prey on duiker (*Cephalophus* spp.; blue duiker, *Philantomba monticola*) at Budongo (Newton-Fisher et al., 2002; Hobaiter et al., 2017), Issa (Ramirez-Amaya et al., 2015), Kibale (Mitani and Watts, 1999; Gilby et al., 2017), Lopé (Gabon; Tutin and Fernandez, 1993), Mahale (Nishida and Uehara, 1983; Hosaka, 2015b; Hosaka et al., 2020), Ndoki (Republic of Congo; Kuroda et al., 1996), and Outamba-Kilimi (Sierra Leone; Alp, 1993, Appendix 1). In contrast, they do not hunt duiker at Taï despite the presence of 6 species (Boesch and Boesch, 1989; Boesch and Boesch-Achermann, 2000).

Based on early Gombe data (Teleki, 1975, Wrangham, 1977), Butynski (1982a; cf. Hill, 1982) concluded that chimpanzees spent ≤1% of feeding time eating meat. Current data show that estimates for mean annual meat intake vary among sites by a factor of about 8 for males and 4 for females (Table 3). Caveats apply: we lack precise data on prey body masses, on the proportion of edible tissue, and on prey energy value and nutrient content; feeding rate estimates have unknown accuracy; and much interindividual and temporal variation exists. However, this does not preclude cross-site comparison.

Table 3
Estimates of per capita chimpanzee meat intake rates and overall annual per capita intake. Most values are wet-weight mass estimates; Boesch (1994a) estimated energy intake rates for Taï and Kasekela. Mahale researchers assumed males and females had equal access to meat during meat-eating sessions (Hosaka et al., 2020).

Site	Adult male	Adult female	Notes	Reference
Mahale	38 g/day, 13.9 kg/year	37g/day, 13.5 kg/year[a]	Alpha male 194 g/d, 70.8 kg/y	Hosaka et al. (2020)
Ngogo	15–20 kg/year	3.7 kg/year	Male range 0–60 kg/y	Watts and Mitani (2002)
Tai	186 g/day, 67.9 kg/year	25g/day, 9.1 kg/year	Much temporal variation	Boesch and Boesch (1989)
Taï	1250–5166 kJ/h		Hunters only; others less	Boesch (1994a)
Taï	3908 kJ/bout		Mean value for hunters	Boesch (1994a)
Kasakela	25 kg/year	8 kg/year	Wide variation	Wrangham and Bergmann-Riss (1990)
Kasekela	81–4560 kJ/h		Hunters only; others less	Boesch (1994a)
Kasekela	1.16 kg/bout		Mean per bout	Gilby (2006)

[a] Hosaka et al. (2020) assumed that females and males present at hunts had equal access to meat.

For example, long-term data from Kanyawara and Kasekela showed red colobus capture rates to be about 8 times higher at Kasekela (Table 3; Gilby et al., 2015) and captures for all prey species combined were 3times higher at Kasekela (Gilby et al., 2017). The average numbers of adults in each community were similar during the period of data collection (11 males and 17 females at Kasekela; 12 males and 16 females at Kanyawara; Gilby et al., 2017); if average prey size was also similar, per capita meat availability would also have been much higher at Kasekela. Capture rates at Mitumba (also in Gombe) were intermediate for red colobus (Table 3), and for overall prey capture rates, there were nearly identical to those for Kanyawara (Gilby et al., 2017). However, the Mitumba community had an average of only 3 adult males and 8 adult females (Gilby et al., 2017); thus, per capita meat availability there would have been higher than at Kanyawara.

Ngogo chimpanzees make kills in over 80% of red colobus hunts, capture over 3 prey per successful hunt, and succeed in more than 50% of hunts of other monkey species (Watts and Mitani, 2015; unpubl. data). Between 1995 and 2001, estimated mean prey biomass per successful red colobus hunt was 15.6 kg; the maximum available from one hunt exceeded 50 kg and estimated per capita availability to males present at hunts averaged nearly 1 kg and was as high as 3 kg. The most successful males harvested 40–60 kg of red colobus (not all edible) annually, an order of magnitude more than the least successful (Watts and Mitani, 2002). Meat accounted for 12.3% of feeding time at Ngogo during 5 months in 2002 (Watts, unpubl. data) when hunting was particularly common (46 successful hunts, 169 prey captures; Watts and Mitani, 2015). Per capita meat availability during that time was 2–3 kg per month. However, individuals were present at varying numbers of hunts, not all those present obtained meat, and those who ate meat did not get equal amounts. Moreover, most individuals eat little or no meat in many months. Per capita meat availability has since decreased as the red colobus population has declined (Watts and Mitani, 2015). Similar variation exists at Kasekela (Stanford, 1998; Gilby, 2006), Kanyawara (Gilby et al., 2008, 2017), Mahale (Uehara, 1997; Hosaka, 2015b; Hosaka et al., 2020), Fongoli (Pruetz et al., 2015), and Taï (Boesch, 1994a; Boesch and Boesch-Achermann, 2000; Fahy et al., 2013; Samuni et al., 2018). Meat intake per hunt at Taï was highest for males who captured prey, intermediate for non-captors who pursued prey, and lowest for individuals who did not hunt, but begged for meat (Boesch, 1994a; Samuni et al., 2018). Adult males eat more meat than adult females (Table 3), and variation among females is considerable (Gilby et al., 2017; Riedel et al., 2019).

Bonobos rarely hunt other primates, which helps explain why they seem generally to hunt less often than chimpanzees. Predation on primates is known only from Lui Kotale (DRC), where bonobos hunt four cercopithecoid species plus Demidoff's bushbabies (*Galagoides demidovii*; Hohmann and Fruth, 2008; Surbeck and Hohmann, 2008; Surbeck et al., 2009), and Isonji (DRC), where Sakamaki et al. (2016) saw a hunt of redtail monkeys (Appendix 1). Wakefield et al. (2019) argued that bonobo-chimpanzee differences are less than assumed and noted that hunting rates for bonobos at Iyema (DRC; Wakefield et al., 2019) and at Lui Kotale (Fruth and Hohmann, 2002) overlap chimpanzee rates (Table 2). However, they acknowledged that their estimate of one duiker hunt per month at Iyema should be treated cautiously because their study lasted only 2 months.

3.4. Nutritional hypotheses

Hypothesis 1: Meat as an energy source Meat can offer considerable energy gains. An adult variegated squirrel (*Sciurus variegatoides*, body mass 428–909 g; Emmons, 1995) could provide 2500 kJ or more to a capuchin if has the meat energy content of North American squirrels (unid. sp.; 120 Kcal/100g; U.S. Department of Agriculture, 2018). A common marmoset could obtain 9–10% of its daily TEE of 213 kJ (Pontzer et al., 2014) from a white-edged tree frog (*Hypsiboas marginatus*; mass = 5–6 g; Giasson and Haddad, 2006, Appendix 1) if the frog's energy content is like that of three frog species eaten by humans in Nigeria (3.75 kJ/g; calculated from Onadeko, 2011). Maximum estimates of the energy that chimpanzees gain during meat-eating bouts are quite high (Boesch, 1994a; Gilby, 2006; SOM Table S1).

However, meat is probably not most important as an energy source. Some small primates meet much of their energy needs from ASFs, but mostly or entirely by eating invertebrates (Hamilton and Busse, 1978). Occasional predation on vertebrates like tree frogs would provide supplements. Energy returns from ASFs come mostly from invertebrates even for baboons (Hamilton and Busse, 1978; Rhine et al., 1986) and white-faced capuchins (Rose et al., 2003). Sex and individual differences in access to meat are pronounced in baboons. During Hausfater's (1976) study of yellow baboons, 5 males accounted for 73% of all meat eating, and males hunted more often and ate more meat than females among olive baboons at Gilgil (Harding, 1975; Strum, 1981) and Mikumi (Rhine et al., 1986) and chacma baboons at Moremi (Hamilton and Busse, 1978). Males usually stole all but small prey from females, who mostly acquired meat only after males discarded it (Hausfater, 1976; Hamilton and Busse, 1982). Males made all captures during Goffe and Fischer's (2016) study of Guinea baboons, although females obtained some meat from them.

White-faced capuchin and baboon data do not support the specific energy shortfall hypothesis. Fruit, the main energy source, was relatively scarce during dry seasons at Santa Rosa during Rose's capuchin (1997) study. Meat intake was highest then, but because of the seasonal availability of coati pups; with body masses of c. 1–200 g (MacDonald, 2006), they probably gave only modest energy returns (Rose, 1997). Bergstrom et al. (2019) implicitly tested the hypothesis with more recent Santa Rosa data. Fruit availability

was lowest in the early rainy season, during part of which invertebrates contributed most food calories. All females needed to eat invertebrates to meet dry season energy and protein requirements, and lactating females would not have met requirements in most months without eating them. Meat-eating was apparently too uncommon to merit mention as an alternative energy source. In reviewing baboon data, Mitani and Watts (2005) found that hunting rates did not show consistent seasonal variation. At Gilgil, where rates were highest, prey captures occurred throughout the year, mostly opportunistically (Harding, 1975).

Neither energy intake rate estimates nor data on hunting rates point to energy gain as the explanation for chimpanzee hunting. Boesch (1994a) estimated that per capita net energy gains for male chimpanzees who participated in red colobus hunts ranged from 1250 to 5166 kJ/h at Taï and 86–4245 kJ/h at Kasekela, depending on the number of hunters. Adding estimated pursuit costs to convert these to gross energy values gives ranges to 1938–5854 kj/h for Taï and 1176–4560 kJ/h for Kasekela and a value of 3908 kj for an average bout at Taï (SOM S1). Applying mean values for wolf prey (SOM S1) to Gilby's (2006) wet-weight intake estimates for Kasekela yields a gross energy intake estimate of 5512 kJ per average meat-eating bout (SOM Table S1).

How easily could chimpanzees match these values by eating PSFs? Based on data on the energy values of chimpanzee PSFs (SOM Table S2), up to 1.5 kg of fruit, 2.7 kg of figs, or 1.9 kg of leaves would be needed to equal the gross energy intake from the average meat-eating bout in Gilby's (2006) sample, and up to 1.3 kg of fruit, 2.9 kg of figs, and 1.9 kg of leaves to equal intake from the average bout at Taï (Boesch, 1994a; SOM Table S3). Replacing 1 hour of meat eating would require c. 1.4 kg of fruit, 2.5 kg of figs, and 1.8 kg of leaves at Taï and 1.1 kg of fruit, 2.1 kg of figs, and 1.5 kg of leaves at Kasekela for the highest per capita meat intake rates, although lower intake rates would require correspondingly less time (SOM Table S3). Uwimbabazi et al. (2018) estimated that chimpanzees at Kanyawara gain 2484 kJ/h from fruit, 3195 kJ/h from figs, and 1560 kJ/h from leaves, much less than Gilby's estimated mean return rate and below Boesch's maximum estimate, but well above Boesch's minimum. Also, only about 270 kJ would be available from a 50 g meat scrap, smaller than the 'typical' scrap at Kasekela (Tennie et al., 2014). PSFs could easily replace this; likewise, they could easily replace the mean daily accessible energy available from meat at Gombe, Taï, Ngogo, and Mahale (SOM Table S3; Fig. 2). Replacement values range from a minimum of 8 g of fruit (females at Ngogo) to 287 g of figs (males at Taï). Males could have met about 44% and 31% of estimated TEE from an average meat-eating bout at Kasekela (Gilby, 2006) and Taï (Boesch, 1994a, b), respectively (SOM Table S3). However, meat-eating bouts like this are intermittent, and on a mean daily basis, meat would have provided only about 1–5% of TEE for males and 0.3–0.8% for females at these and several other research sites (SOM Table S1).

Tool-assisted nut-cracking provides as much as 14,441 kJ, per day at Taï (Boesch and Boesch-Achermann, 2000); depending on the amount of time needed to extract this much energy, expected energy return rates from nut-cracking might exceed the maximum return from hunting, making nut-cracking a better source of energy when nuts are available. The estimated maximum caloric returns from an ant-dipping bout at Kasekela was 247 kJ (O'Malley and Power, 2014). This is far less than Gilby's estimated return from an average meat-eating bout but more than mean daily intake values for females at Ngogo and Taï and close to the estimated gross energy value of a 50 g meat scrap (above). Also, insectivory is less risky, more reliable, and has lower energy costs (McGrew, 1979; Tennie et al., 2014).

Chimpanzee data do not support the energy shortfall hypothesis. Red colobus hunts are most common during rainy seasons at Taï

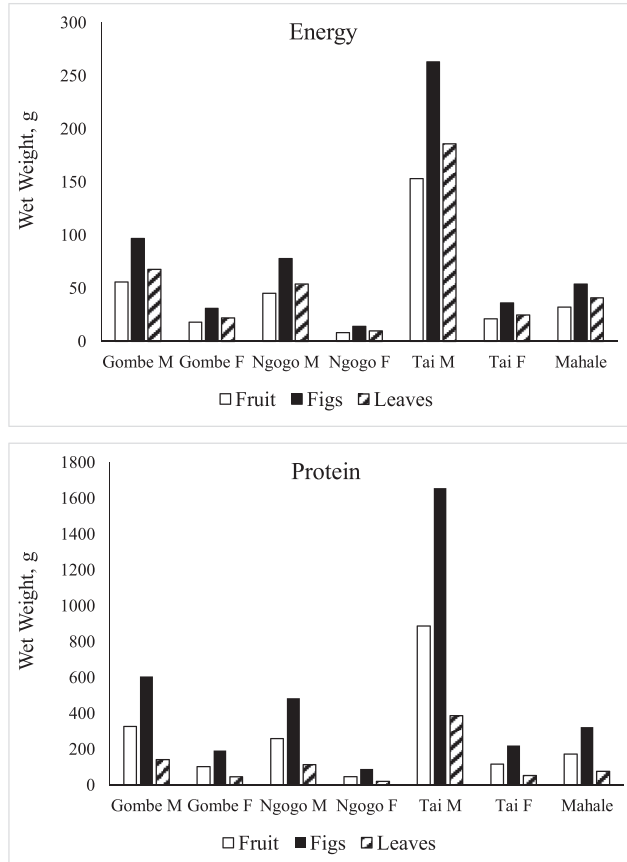

Figure 2. Wet-weight amounts of fruit, figs, and leaves needed to equal the mean amount of energy (upper panel) and protein (lower panel) provided per day by meat at 4 chimpanzee research sites. M = male; F = female (Mahale data for males only).

(Boesch and Boesch, 1989; cf.; Boesch and Boesch-Achermann, 2000). Rainfall and fruit availability are not significantly related, and fruit abundance is relatively high during rainy-dry season transitions (Polansky and Boesch, 2013; Janmaat et al., 2016). Chimpanzees at Kanyawara were more likely to hunt red colobus on encounter during periods when fruit other than figs accounted for at least 40% of feeding time for at least 2 consecutive months, an animal-centered measure of fruit availability, than when non-fig fruit was less abundant (Gilby and Wrangham, 2007). Gilby and Wrangham argued that the chimpanzees could afford risk-prone foraging (hunting) to gain nutrients other than energy when they could easily obtain energy from fruit. Hunting rates at Kasekela also were positively related to fruit availability, although because both were positively related to male party size (Gilby et al., 2008). At Ngogo, monthly hunting rates increased significantly in association with another animal-centered measure of fruit abundance—the proportion of monthly feeding time devoted to the top fruit species in the diet (a relationship driven by red colobus hunting; Watts and Mitani, 2015). Most hunts occurred during 'hunting patrols', when the chimpanzees searched for red colobus groups for up to several hours and fed little on PSFs (cf. Mitani and Watts, 1999; Watts and Mitani, 2002; Samuni et al., 2018). Engaging in such searches should not impose net costs despite their uncertain returns: individuals gain energy from any meat they consume and can easily gain it from fruit before and after patrols and hunts (Watts and Mitani, 2002, 2015).

Hypothesis 2: Meat as a protein source Small vertebrate prey provide valuable protein supplements for small-bodied primates that, like white-faced capuchins (Bergstrom et al., 2019), get most

of their protein from invertebrates. A white-edged tree frog would give a common marmoset about 1 g of protein. At the opposite extreme, the 1.16 kg of colobus meat consumed during an average bout at Kasekela (Gilby, 2006) would yield about 229 g of protein if its protein content were the same as the mean value for wolf prey (SOM Table S1). Based on PSF nutritional data (SOM Table S2), a chimpanzee would need up to 10.4 kg of fruit, 18.6 kg of figs, and 3.9 kg of leaves to gain the same amount of total protein (SOM Table S3). Error notwithstanding, providing the protein sometimes available from meat clearly would take large volumes of PSFs. Males at Taï had the highest estimated mean daily protein intake, at 24.8 g/day (SOM Table S1; calculated from data in Boesch and Boesch, 1989), enough to give an average 46.3 kg male (Smith and Jungers, 1997) about 0.54 g protein/kg/day. To get that from PSFs would require about 900 g of fruit, 1.7 kg of figs, or 400 g of leaves (Fig. 2; SOM Table S3). Given wet weight intake rates for PSFs calculated from Uwimbabazi et al. (2018), males could have reached these values by eating fruit for 73 min, figs for 81, or leaves for 34. Estimated mean daily meat intake values (Table 3), and thus PSF equivalence values (Fig. 2, SOM Table S3), were much lower for females at Taï and were lower for both sexes at other sites. Ngogo females, for whom values were lowest, could have matched their mean daily total protein intake from meat with about 47 g of fruit, 90 g of figs, and 21 g of leaves. One hundred seventy-nine g of leaves would have about 11 g of protein, the same amount as in a 50 g meat scrap. Mean daily values mask high daily variance, but matching total protein intake from meat by eating PSFs should generally not be difficult. N'Guessan et al. (2009; estimated from Figure 4) used data on feeding rates and PSF nutrient content to infer that female chimpanzees at Taï averaged about 60–200 g in total daily protein intake and males about 60–160 g. They did not specify how much protein came from meat, and these numbers do not take digestibility of PSFs or bioaccessibility into account. Still, although they indicate that individuals who have access to large amounts of meat after hunts can more than match total daily protein intake from PSFs, Milton's (1999) conclusion that chimpanzees get most protein from PSFs seems secure.

A similar conclusion holds for baboons and capuchins. A white-faced capuchin could acquire ≥50 g of protein by consuming a variegated squirrel, which presumably contains about 21.3 g protein/100 g wet weight (U.S. Department of Agriculture, 2018). Coati pups and other small vertebrate prey would provide smaller amounts. However, as Rose (1997) argued and Bergstrom et al. (2019) implied, protein from vertebrate prey probably supplements more dependable protein intake from invertebrates and PSFs. Data on the nutritional content of meat from rabbits (*Oryctolagus cuniculus*), wildebeest (*Connochaetes taurinus*), steenbok (*Raphisteros capestris*), and duiker (Kyriacou et al., 2016; SOM S1) show that baboons sometimes gain substantial amounts of protein by preying on these species, but prey captures are infrequent.

Predation on rats by pig-tailed macaques (Ruppert et al., 2018; Holzner et al., 2019) is an interesting case. Holzner et al. (2019) estimated that a group of 40–50 macaques killed 3000 rats per year. Presumably not all group members catch rats. If mean body mass for the rats is about 140 g, about 67% of which is edible (SOM S1), and their meat has about 24.5 g protein/100 g (the mean for 12 wild rodents eaten by people in Africa; Ntiamoa-Baidu, 1997), 30 macaques would harvest about 9.4 kg of edible meat per capita annually and about 6 g gross protein per day. These values exceed those estimated for female chimpanzees at all sites listed in Table 3 and SOM S1 and for males at Mahale.

Hypothesis 3: Vitamins, minerals, and micronutrients Vitamins, minerals, and micronutrients (collectively referred to as 'micronutrients', below) are needed for myriad processes, including immune function, brain growth, cognitive development, and cell division (Bailey et al., 2015; Stammers et al., 2015). Some are widely available in PSFs, especially in certain plant parts, and intakes from PSFs sometimes equal or exceed recommended daily allowances (Milton, 2003b; Cancelliere et al., 2014). But, ASFs are often rich in vitamins A, K, B_6, and B_{12}, calcium, phosphorus, sodium, potassium, iron, selenium, zinc, and other micronutrients generally scarce in plants (Milton, 2003b; Payne et al., 2016; Phillips et al., 2015). For example, sodium, iron, zinc, and copper in 8 African bovid species (Kyriacou et al., 2016; FAO, 2019) and wolf ungulate prey (Bosch et al., 2016) are higher than those in fruit and leaves eaten by diademed sifaka (*Propithecus diadema*; Irwin et al., 2017) and mountain gorillas (*Gorilla beringei*) in Bwindi (Cancelliere et al., 2014, Fig. 3). Phosphorus values were also higher for ungulates (springbok: 0.71%; FAO, 2019: wolf prey: mean = 0.74%; Bosch et al., 2016) than for sifakas and mountain gorilla PSFs (range = 0.17–0.34% for fruit, herb leaves, and tree leaves; Cancelliere et al., 2014; Irwin et al., 2017), as were manganese values (797 ppm for wolf prey vs. 79–275 ppm for sifaka and gorilla foods).

Such differences underlie Tennie et al.'s (2009) meat scrap hypothesis: needs for micronutrients, not energy or protein, motivate group hunts by chimpanzees. Support for the hypothesis included (Tennie et al., 2009): (1) per capita meat availability at Kasekela decreased with the number of individuals present per hunt and was independent of the number of hunters there (cf. Gilby et al., 2006) and at Ngogo (cf. Watts and Mitani, 2002); (2) net caloric gain was independent of hunting group size (cf. Boesch, 1994a); and (3) the probability that individuals obtained some meat increased with hunting group size (also the case at Ngogo; Watts and Mitani, 2002). The model's logic also applies to the questions of why omnivores such as chimpanzees, baboons, and capuchins hunt vertebrates at all (Tennie et al., 2009; Teleki, 1973, 1975; Hamilton and Busse, 1982; Milton, 2003a, b) and why chimpanzees and other large-bodied primates engage in insectivory (Tennie et al., 2014).

Certain tissues can be particularly important micronutrient sources. Liver copper content exceeded that in muscle in all 6 wild African bovid species for which Kyriacou et al. (2016) had data, and liver iron content was higher in 5. Copper and iron contents were considerably higher in liver than skeletal muscle in wolf ungulate prey; stomach, lung, and spleen tissue had considerably more iron, and lungs, stomach, and kidneys had considerably more sodium (Bosch et al., 2016). Brain and liver are particularly valuable sources of long-chain polyunsaturated fatty acids needed for brain growth,

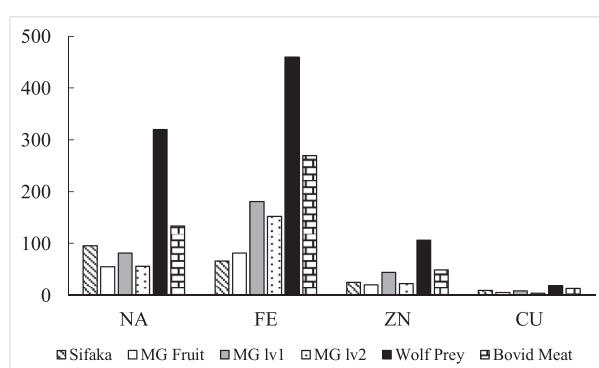

Figure 3. Contents (PPM) of sodium, iron, zinc, and copper in plant-source foods of diademed sifakas (*Propithecus diadema*; Irwin et al., 2017), mountain gorillas (*Gorilla beringei* [MG]; Cancelliere et al., 2014), ungulate prey of wolves (Bosch et al., 2016), and 9 species of African bovids (Kyriacou et al., 2016; FAO, 2018). Sifaka data are for fruit and foliage combined. MG lv1 = leaves from trees; MG lv2 = leaves from terrestrial vegetation.

brain metabolism, and retinal development and of their omega-3 and omega-6 precursors (Kyriacou et al., 2016). Chimpanzees eagerly eat internal organs, including colobine stomach and intestinal contents, which they seem to prefer to other tissues (Gilby et al., 2017; pers. observ.). The importance of long-chain polyunsaturated fatty acids could explain why Gombe chimpanzees usually eat the brains of infant red colobus, to which they can gain direct access by biting into the skulls, before other carcass parts (Gilby and Wawrzyniak, 2018).

The meat scrap hypothesis helps explain the eagerness of chimpanzees, bonobos, baboons, and capuchins to obtain meat from others (Tennie et al., 2009). Secondary meat acquisition is common in *Pan* (e.g., Mitani and Watts et al., 2012; Fruth and Hohmann, 2002; Samuni et al., 2018); this helps explain why the number of chimpanzees who ate meat per hunt increased with hunting party size at Kanyawara and Ngogo (above) and with carcass size and total meat availability at Ngogo (unpubl. Ngogo data). Theft and peaceful transfers occur in capuchins, but retrieval of dropped/discarded meat is the most common means of secondary acquisition, whereas theft and retrieval of dropped or discarded meat are the main modes of secondary acquisition in baboons (Harding, 1975, 1975; Strum, 1981; Hamilton and Busse, 1982; Goffe and Fischer, 2016). The mean number of individual capuchins who controlled access to squirrel and coati carcasses was 1.68 at Santa Rosa and 2.98 Lomas Barbudal. One to 16 baboons fed on carcasses at Gilgil; up to 7 fed directly (often on discarded carcasses), and up to 11 ate fallen scraps; 51% of individuals who ate meat only got scraps (Strum, 1981). Transferred meat sometimes provides substantial protein and calories but always offers micronutrients.

4. Discussion

4.1. Implications for human evolution

The last common ancestor of chimpanzees, bonobos, and humans presumably hunted and ate meat (Wood and Gilby, 2017), although baboons might be a better model than chimpanzees for making inferences about hominin diets starting 3 to 4 Ma because of their greater reliance on C_4 foods and their great ecological flexibility (Codron et al., 2008; Lee-Thorp et al., 2010). Some nonhuman primates sometimes gain protein and energy bonanzas from large meat meals. Some chimpanzees eat substantial meat annually; whether this affects fitness is unknown. However, meat may typically be most important as a source of micronutrients uncommon in PSFs. When ASFs are consistently major protein and energy sources, those foods are usually invertebrates (e.g., for white faced capuchins: Bergstrom et al., 2019; for primates that rely heavily on plant exudates, which have little protein: Hladik, 1979). Continued collection of detailed data on the nutritional composition of PSFs and ASFs (including specific parts of mammalian prey carcasses) should help identify supplementary and/or complementary nutritional contributions of ASFs. In contrast, vertebrate ASFs were substantial components of all known hunter-gatherer diets (Cordain et al., 2000; Kelly, 2007), contributed substantially to energy intake and often provided the majority of dietary calories, and were typically the major source of protein (Cordain et al., 2000; Kaplan et al., 2000; Kelly, 2007; Wood and Gilby, 2017). The literature spanning the gap between the last common ancestor of chimpanzees and humans and the ecology of modern hunter-gatherers is too vast for review, but I will briefly discuss several issues that the nonhuman primate data highlight.

Detailed quantitative data on ASFs in hunter-gatherer diets are limited (Crittenden and Schnorr, 2017). In reviewing those from 9 societies, Kaplan et al. (2000) estimated that meat provided a mean of 52.7% of food calories (range = 25.8–78.8%) plus most protein. Estimated individual meat intake varied from 260 to 1360 g per day. The lowest value exceeds the highest estimated daily mean for chimpanzees (males at Taï; Boesch and Boesch, 1989) but only by a factor of 1.4. Meat contributed more to hunter-gatherer diets in the temperate zone and Arctic than in the tropics (Cordain et al., 2000; Kelly, 2007); Wood and Gilby (2017) thus limited their comparisons of chimpanzees and hunter-gatherers to data from 6 African societies, in which meat provided 9%–50% of dietary calories. If the estimated contributions of meat to chimpanzee TEE (above) are roughly accurate, some Taï males came close to the lowest value, and meat intake by some males during periods of elevated hunting rates there and at Ngogo might temporarily have exceeded this value. However, the lowest hunter-gatherer value (the Efe) in Wood and Gilby (2017) was less than half the others, and other chimpanzee values are much lower than Taï values. The point stands that recent tropical hunter-gatherers ate much more meat than any nonhuman primates do (Wood and Gilby, 2017).

Moreover, meat started to assume greater nutritional importance for some hominins (presumably members of *Homo*, but perhaps other genera) by at least Ma and increased use of bone marrow might have marked a fundamental break with chimpanzee-like or baboon-like faunivory well before then (Thompson et al., 2019). Possible indications of a significant increase in meat eating occur at the earliest sites with uncontested evidence of carcass processing using Oldowan technology (Bouri: de Heinzelin et al., 1999; several Gona sites: Semaw et al., 2003). The small Bouri and Gona samples leave inferences of possible early carcass access (Dominguez-Rodrigo et al., 2005) open to question but indicate that at least the tool users acquired meat and marrow from large mammals (Bunn, 2007). Larger samples at later Oldowan sites (e.g., FK22 Zinj at Olduvai; FxJj 50 at Koobi Fora; the ST complex at Peninj) make a compelling case that some Oldowan hominins (presumably members of *Homo*) had regular access to intact medium-to-large ungulate carcasses, butchered meat-rich skeletal elements, and carried substantial meat from butchery sites to other places for consumption (Dominguez-Rodrigo and Barba, 2006; Bunn, 2007; Dominguez-Rodrigo, 2012; Lupo, 2012; Pickering and Bunn, 2012). They gained early access to at least some carcasses, perhaps via confrontational scavenging but possibly also via stalking and ambushed prey, perhaps using wooden spears made with stone tools (Bunn and Pickering, 2010; cf.; Pickering and Bunn, 2012) The prey mortality profile from the 1.8 Ma site of FLK Zinj matches that expected from ambush hunting (Bunn and Gurtov, 2014). Importantly, abundant remains of medium-sized bovids at Kanjera show signs of butchery consistent with early carcass access, but few or no signs of carnivore damage, showing that at least one hominin taxon regularly engaged in carnivory by 2 Ma (Ferraro et al., 2013). Later Olduvai BK sites give evidence for systematic butchery of complete or nearly complete ungulate carcasses with a size range from gazelle to elephants at 1.34 Ma (Dominguez-Rodrigo et al., 2014). The larger carcasses would have provided meat bonanzas to large hominin groups (Dominguez-Rodrigo et al., 2014).

Wood and Gilby (2017) made the important point that most prey killed by the African hunter-gatherers in their sample weighed $\leq$10 kg and were thus within the size range of chimpanzee prey. However, they added that hunters sometimes killed prey much larger than themselves, as temperate and Arctic hunter-gatherers did routinely. This is a fundamental characteristic of the human predatory complex (Thompson et al., 2019). Like all nonhuman primates other than *T. bancanus*, chimpanzees only take prey much smaller than themselves. Adult male chimpanzees are 4 to 5 time larger than their largest prey (adult male black and white colobus, *C. guereza* and *C. angolensis*; body masses c. 9–13 kg; adult male red

colobus, c. 9–12 kg), and adult females and older adolescents are also considerably larger than any prey. Moreover, chimpanzees mostly capture immature individuals. At birth, duiker are c. 10% of adult body mass, which varies from 4 to 9 kg (blue duiker) to c. 24 kg (e.g., Bay duiker *Cephalophus dorsalis*), a species on which chimpanzees and bonobos prey; they reach 20% of adult mass within a month (Kendall, 2000). Predation is typically restricted to infants and juveniles, although Hohmann and Fruth (2008) documented one capture of a mature bay duiker (15–24 kg) by bonobos (mean adult body mass = 34.3 kg for females, 42.7 kg for males; Zihlman and Bolter, 2015). Adults of some artiodactyls (e.g., bushbuck; bush pigs, *Potomochoerus larvatus*) on which mandrills, baboons, and chimpanzee prey (Appendix 1) are larger than the primates, but only infants and/or juveniles are prey. As Werdelin (2017) pointed out, chimpanzees (and male baboons and mandrills) exceed the 21 kg threshold identified by Carbone et al. (1999) above which carnivorans switch from mainly or exclusively taking prey with body masses far less than their own to capturing prey with equal or larger body masses. Thus 'from the carnivoran perspective, chimpanzees are not carnivores, merely animals that occasionally exploit animal resources' (Werdelin, 2019: 15). Capuchins are also much larger than their typical prey (e.g., *C. capucinus* with variegated squirrels, above; *S. apella*, male body mass c. 3.6 kg, with titi monkeys, male body mass c. 1 kg), although Visalberghi (2019) saw 4 immature capuchins unsuccessfully attack an iguana (*Iguana iguana*) about twice their size.

Documentation of tool-assisted hunting by chimpanzees at Fongoli (Pruetz and Bertolani, 2007; Pruetz et al., 2015) erased a presumed hominin/nonhuman primate dichotomy. But, tool use for prey capture is unusual in nonhuman primates and restricted to extraction (broadly speaking) of small prey, and tool use for carcass processing is nonexistent except for occasional use of sticks to extract bone marrow by chimpanzees (Teleki, 1975). Three cases of stick-assisted prey capture by chimpanzees other than those at Fongoli have been documented, all at Mahale: two extractions of squirrels from tree holes (a red-legged sun squirrel, *Heliosciureus rufobrachum*; Nakamura and Itoh, 2008; *H. rufobrachum* or *Protoxerus stangeri*; Huffman and Kalunde, 1993) and one of a rock hyrax from a small cave (*Heterohyrax brucei*; Nakamura and Itoh, 2008). Bearded capuchins at Serra da Capivara use sticks to extract lizards from rock crevices (Falotico and Ottoni, 2014), and Russon et al. (2014) saw a few cases of tool-assisted fishing by rehabilitant orangutans. Such behavior bears no resemblance to Oldowan hominin use of pounding and cutting tools to process carcasses. But chimpanzee-like nut-cracking presumably preceded use of hammerstones by Oldowan and pre-Oldowan hominins to open marrow-rich bones, a major hominin innovation in carcass processing (Thompson et al., 2019); likewise, manufacture of wooden tools to extract prey could have been a precursor to cognitively more sophisticated use of stone tools to make wooden hunting tools by Oldowan hominins (Pruetz and Bertolani, 2007; Lemorini et al., 2014).

Hominins dispersed to temperate or largely temperate habitats in Eurasia by 1.85 Ma (Gabunia et al., 2000) and in southwestern France by 1.23 Ma (Huguet et al., 2013). Maintaining a presence in warm temperate southern Europe and, during warmer periods, more northern areas where PSFs would have been seasonally scarce required year-round supplies of meat from small to large mammals and preferential targeting of herd-forming species like bison and horses (Landek-Garriga and Garriga, 2016). Evidence of early access to, and regular butchery of, carcasses of large mammals (e.g., *Cervus, Equus,* and *Bison* sp.) at Untermassfeld (Landek-Garriga and Garriga, 2016), Gran Dolina (Spain: Huguet et al., 2013; Rodriguez-Hidalgo et al., 2017), and multiple sites in southwestern France (Huguet et al., 2013) and the Levant (Speth, 2013), among others, attest to the ability of early *Homo* to engage in flexible and opportunistic predation. Multiple European Middle Paleolithic sites (e.g., level TD10-2 in Gran Dolina: Rodriguez-Hidalgo et al., 2017; Schoningen: Conard et al., 2015) provide evidence for collaborative group hunts of large ungulates that led to multiple kills. Planned, coordinated attacks on large prey have no equivalent in nonhuman primates, nor do those nonhuman primates that hunt other mammals engage in sit-and-wait ambush hunting or slow stalking of prey.

However, in focusing on large prey, we should remember that ethnographically documented hunter-gatherers included "avian game … small ungulates [and] small land fauna (e.g., lizards and rodents)," in their diets (Crittenden and Schnorr, 2017: 87; cf. Bird et al., 2013; Wood and Gilby, 2017). Nor can we assume hunting large game would always have been an efficient tactic, even given suitable technology, because of potentially high pursuit costs and the need to solve collective action problems (Bird et al., 2013). Meat eating by most nonhuman primates primarily involves opportunistic, low-risk predation on small animals (lizards, frogs, small mammals; bird eggs, nestlings, and fledglings) for which pursuit costs are low. Savanna chimpanzees mostly take small prey (above). Focusing on these points might allow progress regarding arguments about the relevance of chimpanzee prey capture tactics to those of bipedal hominins (e.g., Sayers et al., 2012; Stanford, 2012).

The paleoarchaeological record documents a long history of use of small prey, although intensive exploitation was not common until perhaps 70 Ka in Southern Africa (Clark and Kandel, 2013) and until about 30 Ka more globally (Stiner et al., 2000; Shea and Sisk, 2010; Janz, 2016). Data from Southern Africa in particular shows major increases in exploitation of size class 1 bovids (4.5–24 kg), a category that includes steenbok, dikdik, and most duiker species — prey of chimpanzees, bonobos, and/or baboons — and possibly increased exploitation of small mammals, during MIS 4 (Clark and Kandel, 2013). For chimpanzees to capture juvenile red duiker and juvenile or adult blue duiker does not necessarily require fast pursuit (pers. observ.). Nor does seizure of cached ungulate fawns require either prolonged pursuit or tool use. Cached fawns could have been abundantly available to Oldowan hominins during birth seasons in open habitats like Kanjera (Ferraro et al., 2013), provided the hominins had effective search strategies (Oliver et al., 2019), and prey mortality profiles from Kanjera show that Oldowan hominins there frequently captured juveniles of small bovids, behavior that might have required only small chases (Oliver et al., 2019). Likewise, extraction of galagos from tree holes by chimpanzees, with or without tools, does not involve the arboreal agility required to capture red colobus (Pruetz et al., 2015). Chimpanzees and bonobos opportunistically prey on birds, a tactic also documented for the Hadza (Marlowe, 2010) and Kung (Lee, 1968). Observations at Loango (Pika et al., 2019) show that some chimpanzees recognize tortoises as food and even use percussion, albeit not stone tools, to open their carapaces. Evidence for hominin predation on turtles dates to at least 1.95 my (Braun et al., 2010) and comes multiple early and mid-Pleistocene sites, even if it was rare until the mid-Pleistocene (reviewed in Blasco et al., 2016).

Technology allowed early hominins to kill prey larger than themselves. Technological innovation also eventually facilitated greatly intensified resource use and great broadening of diets; together with culturally mediated transmission of technological and ecological knowledge, it facilitated the unprecedented ecological flexibility that allowed *Homo sapiens* to occupy virtually the entire range of terrestrial habitats (Clark and Kandel, 2013; Janz, 2016; Starkovich et al., 2018; Wedage et al., 2019). Technological innovation has contributed to faunivory in nonhuman primates but almost exclusively with regard to harvesting and processing invertebrates. Adaptations for arboreal posture and locomotion found

in chimpanzees, but not in Plio-Pleistocene hominins, have sometimes been cited as reasons why hunting by chimpanzees fails to offer insights into hominin evolution (e.g., Sayers et al., 2012). This argument is sensible incorporated with regard to rapid arboreal pursuit of cercopithecines, but not capture of stationary of slowly moving terrestrial prey or extraction of prey from nests. Humans eventually incorporated fast, difficult-to-capture arboreal prey, like those that chimpanzees pursue, in their diets. Evidence from Sri Lanka now shows that as early as 45,000 Ka, modern humans successfully occupied rainforests, where they engaged in specialized hunting of macaques, langurs, and giant squirrels (Wedage et al., 2019). However, they did this using sophisticated microlithic and bone-tipped projectile technology, without which pursuit of such prey would have been energetically unfeasible, if not impossible (Wedage et al., 2019).

Conflict of interest

The author declares no conflict of interest.

Appendix 1

Consumption of Vertebrate Prey by Nonhuman Primates. Order, family, species, and common name of prey given when available. Under birds (when information available), E = eggs, F = nestlings/fledglings, and A = adults. *Accounts of predation on taxon indicate that it occurs moderately often or commonly.

Predator:	Prey: Class	Order	Family	Species
Strepsirrhini				
Lorisiformes				
Lorisidae				
Euoticus elegantulus	Aves	Passeriformes	Cisticolidae	Grey-backed camaroptera, *Cameroptera brevicaudata*[1]
Galago alleni	Amphibia	Anura		Small frogs, unid. spp.[1]
Galago moholi	Aves	Passeriformes	Monarchidae	African paradise flycatcher, *Tersiphone viridis* E, N[2]
			Turdidae	Kurrichane thrush, *Turdus libonyana* E[2]
Loris lydekkerianus	Reptilia	Squamata	Gekkonidae	Gekkos, unid. sp., "small lizard", unid. spp.[3,4]
Loris tardigradus	Reptilia	Squamata	Gekkonidae	Gekkos, unid. sp., "lizards", unid. spp.[3,4,5]
	Aves			Birds, unid. spp.[5]
Nycticebus coucang	Aves			Birds, unid. sp. E[4,7]
Otolemur crassicaudatus	Reptilia	Squamata		Lizards, unid. spp.[5,8]
	Aves*	Passeriformes	Calyptomenidae	African broadbill, *Smithornis capensis* E[2]
			Monarchidae	African paradise flycatcher, *Terisphone viridis* E, N[2]
			Muscicapidae	Red-capped robin chat, *Cossypha natalensis* E[2]
			Nicatoridae	Eastern nicator, *Nicator gularis* E[2]
			Pycnonodidae	Yellow-bellied greenbul, *Chlorocichla flaviventris* E, N[2]
				Unid. spp.[5]
Perodicticus potto	Aves			Birds, unid. spp. A[1]
	Mammalia	Chiroptera	Pteropodidae	Franquet's epauletted bat, *Epomops franqueti*[1]
Lemuriformes				
Lemuridae				
Cheirogaleus medius	Reptilia	Squamata		Lizards, unid. spp.[9]
Eulemur fulvus	Aves	Passeriformes	Monarchidae	Madagascar paradise flycatcher, *Terisphone mutata*[10]
	Mammalia	Primates	Lemuridae	Ringtailed lemurs, *Lemur catta*[11,12]
Microcebus berthae	Reptilia	Squamata	Chamaeleonidae	Chameleons, unid. spp.[13]
			Gekkonidae	Gekkos, unid. spp.[13]
Microcebus murinus	Amphibia	Anura		Frogs, unid. spp.[5,14]
	Reptilia	Squamata	Chamaeleonidae	Chameleons, unid. spp.[5,9,13,14]
			Gekkonidae	Gekkos, unid. spp.[5,13,14]
Mirza coquereli	Amphibia	Anura		Frogs, unid. spp.[15]
	Reptilia	Squamata	Chamaeleonidae	Chameleons, unid. sp.[5,9]
		Squamata		Lizards, unid. spp.[15]
	Aves			Birds, unid. spp. A, E[15]
	Mammalia	Rodentia		Small rodents, unid. spp.[15,16]
		Primates	Cheirogaleidae	Mouse lemurs, *Microcebus murinus*[15,16]
Haplorrhini				
Tarsiiformes				
Tarsiidae				
Tarsius bancanus	Teleostei	Actinopterygii		Fish, unid. spp.[17]
	Reptilia*	Squamata	Elapidae	Malaysian blue coral snake, *Calliophis intestinalis*[17,18]
	Squamata			Lizards, snakes, unid. spp.[5,17,18,19,20,21]
	Aves*	Coraciiformes	Alcedinidae	Kingfishers, unid. sp.[18]
		Passeriformes	Sylviidae	Warblers, unid. sp.[18]
			Pittidae	Pitta, *Pitta* sp.[18]
				Birds, unid. spp.[16,17,18,19,20,21]
	Mammalia	Chiroptera	Pteropodidae	Horseshoe bat, *Taphozus* sp.[18]
				Lesser short-nosed fruit bat, *Cynopterus brachiotys*[18]
				Spotted-winged fruit bat, *Balionycteris maculata*[18]
Anthropoidea				
Platyrrhini				
Atelidae				
Alouatta caraya	Aves	Galliformes	Phasianidae	Domestic chicken, *Gallus gallus*, A[22]
Alouatta palliata	Aves			unid. sp. N[23]
Alouatta seniculus	Reptilia	Squamata	Iguanidae	Green iguanas, *Iguana iguana*[24]
Lagothrix flavicauda	Reptilia	Squamata	Colubridae	unid. sp.[25]
Lagothrix lagotricha	Amphibia	Anura	Hyelidae	Giant gladiator tree frog, *Boana boans*[26]

(*continued on next page*)

Appendix 1 (*continued*)

Predator:	Prey: Class	Order	Family	Species
		Anura		Frogs, unid. spp.[27]
	Mammalia	Rodentia		Rodents, unid. spp.[27]
Callitrichidae				
Callimico goeldii	Amphibia	Anura*		Frogs and toads, unid. spp.[28]
	Reptilia	Squamata*		Small lizards, unid. spp.[28]
	Aves*			Birds, unid. spp. E, N[28]
Callithrix aurita	Aves			Birds, E[29]
Callithrix flaviceps	Amphibia	Anura*		Frogs, unid. spp.[31,32]
	Reptilia	Squamata*		Lizards, unid. spp.[32]
	Aves			Birds, unid. spp. E, N[31,32]
Callithrix geoffroyi	Amphibia	Anura*		Frogs, unid. spp.[33]
	Reptilia	Squamata*		Small lizards, unid. spp.[33]
Callithrix humeralifer	Amphibia	Anura		Frogs, unid. spp.[31]
	Reptilia	Squamata		Lizards, unid. spp.[31]
	Aves	Passeriformes	Thamnophilidae	Ant wren, *Myrmotherula cyanocphala*[34]
			Thraupidae	Tanger, *Tangara* sp.[34]
				Birds, unid. spp. A[31]
Callithrix jacchus	Amphibia	Anura	Hylidae*	White-edged tree frog, *Hypsiboas albomarginatus*[35]
				Frogs, unid. spp.[35,36,37]
	Reptilia	Squamata	Gekkonidae*	Brazilian gecko, *Phyllopiza pollicaris*[38]
				Small lizards, unid. spp.[35,36,37,39,40,41,42,43,44]
	Aves*	Apodiformes	Trochilidae	Swallow-tailed hummingbird, *Eupetomena macroura* E[45]
				Reddish hermit, *Phaethornis ruber* E[45]
		Caprimuligiformes	Capromulgidae	Paroque, *Nyctidromis albicollis* E[45]
		Colombiformes	Columbidae	*Columbia livida domestica* E,F[43]
				Plain-breasted ground dove, *Columbina minuta* E[45]
				Ruddy ground dove, *Columbina talpacoti* E[45]
		Passeriformes	Cotingidae	White-winged becard, *Pachyramphus polychopterus* E[45]
			Thamnophilidae	White-fringed ant wren, *Formicivora grisea* E[45]
			Thraupidae	Burnished buff tanager, *Tangara cayana* E[46]
C. jacchus, cont.			Turdidae	Pale-breasted thrush, *Turdus leucomelas* E[46]
				Ruddy ground thrush, *Turdus rufiventris* E,F[43]
			Tyrannidae	Yellow-Bellied elaena, *Elania flavogaster* E[45]
				Great kiskadee, *Pitanga sulphuratus* E[45]
				Birds, unid. spp. E, N[36,37,38,44]
	Mammalia	Didelphimorphia	Didelphidae	Woolly mouse opossum, *Micoureus demerarae*[41]
				Gray short-tailed opossum, *Monodelphis domestica*[41]
		Rodentia	Echimyidae	Tree rat, *Makalata didelphoides*[41]
				Rodent, unid. sp.[42]
Callithrix jacchus -pencillata	Aves	Columbiformes	Columbidae	Eared dove, *Zenaida auriculata* E[47]
Callithrix kuhli	Amphibia	Anura		Frogs, unid. spp.[32]
	Reptilia	Squamata		Lizards, unid. spp.[32]
	Aves			Birds, unid. spp. A[32]
Callithrix penicillata	Amphibia	Anura	Hylidae*	Lund's tree frog, *Boana lundii*[48]
				White-edged tree frog, *Hypsiboas albomarginatus*[49]
				Frogs, unid. sp.[50]
	Reptilia*			Lizards, unid. sp.[50]
	Aves*	Passeriformes	Columbidae	Ruddy ground dove, *Columbina talpacoti* A[51]
			Icteridae	Variable oriole, *Icterus pyrrhopterus*, F[52]
				Unid. spp. E[52]
	Mammalia	Didelphimorphia	Didelphidae	Agile opossum, *Gracilianus agilis*[53]
		Rodentia	Muridae	Mouse, *Mus musculus*[51]
Cebuella pygmaea	Amphibia	Anura	Aromabatidae	3-striped rocket frog, *Allobates cf. trilineatus*[54]
	Reptilia	Squamata	Dactyloidea	Anoles, *Anolis* sp.[54]
			Tropiduridae	Blue-lipped tree lizard, *Plica umbra*[55]
Leontopithecus chrysomelas	Amphibia	Anura*		Frogs, unid. spp.[56,57]
	Reptilia	Squamata*	Dactyloidea	Anoles, *Anolis* sp.[57]
			Tropiduridae	Amazon lava lizard, *Tropidurus torquatus*[57]
				Lizards, small snakes, unid. spp.[56,57]
	Aves			Birds, unid. spp. E[57]
Leontopithecus chrysopygus	Amphibia	Anura*		Frogs[58]
	Reptilia	Squamata*	Dactyloidea	Anoles, *Anolis* sp.[57]
			Tropiduridae	Amazon lava lizard, *Tropidurus torquatus*[57]
Leontopithecus rosalia	Amphibia	Anura		Frogs, unid. spp.[57,59,60,61]
	Reptilia	Squamata		Lizards, unid. spp.[57,59,60,61]
			Tropiduridae	Amazon lava lizard, *Tropidurus torquatus*[57]
			Dactyloidea	Anoles, *Anolis* sp.[57]
	Aves	Passeriformes	Turdidae	Rufous-bellied thrush, *Turdus rufiventris* E[57]
Saguinus fuscicollis	Amphibia	Anura*		Frogs, toads, unid. spp.[62,63,64,65,66,67,68]
	Reptilia	Squamata*		Lizards, unid. spp.[62,63,64,65,66,67,68]
	Aves			Birds, unid. spp. N[65,66,69]
Saguinus imperator	Amphibia	Anura		Frogs, unid. spp.[63]

Appendix 1 (*continued*)

Predator:	Prey: Class	Order	Family	Species
	Reptilia	Squamata		Lizards, unid. spp.[63]
Saguinus mystax	Amphibia	Anura	Hylabatidae*	Tree frogs, *Hyla* spp.[69]
				Tree frogs, *Phyllomedusa* spp.[69]
				Slender-legged tree frogs, *Osteocephalus* spp.[69]
				Frogs, unid. spp.[65,67,70]
	Reptilia	Squamata*		Lizards, unid. spp.[65,67,69,70]
Saguinus oedipus	Amphibia	Anura		Frogs, unid. spp.[71,72,73]
	Reptilia	Squamata		Lizards, unid. spp.[71,72,73]
Cebidae				
Cebus albifrons	Amphibia	Anura*		Frogs, unid. spp.[63]
	Reptiles	Squamata*		Lizards, unid. spp.[63]
	Aves*	Passeriformes	Icteridae	Yellow-rumped cacique, *Cacicus cela*[74]
	Mammalia*	Rodentia	Muridae	Rats, unid. sp.[63]
		Didelphimorphia	Didelphidae	Opposums, unid. sp.[63]
Cebus capucinus	Amphibia	Anura*		Frogs, unid. spp.[75,76,77,78]
	Reptilia	Squamata*	Iguanidae	Spiny lizards, *Ctenosaura simila*[78]
			Anolidae	Anoles, *Anolis* spp.[78]
				Lizards, unid. spp.[75,76,77,78]
	Aves*	Columbiformes	Columbidae	Ruddy ground dove, *Columbina talpacoti* A, N[77,78]
		Cuculiformes	Cuculidae	Greater ani, *Crotophaga major* E,N E[79]
		Galliformes	Cracidae	Great curassows, *Crax rubra* E[77,78]
Cebus capucinus				Crested guans, *Penelope purpurescens* E[77,78]
		Passeriformes	Corvidae	White-throated magpie, *Callocitta formosa* N[77,78]
			Troglodytidae	Rufous-naped wren, *Camplorhynchus rufinucha*[78]
				Banded-wrens, *Thryothorus pleurostictus* E, N[77,78]
				Unid. sp.[78]
		Psittaciformes	Psittacidae	Scarlet macaw, *Ara macau*[80]
				Parrots, unid. sp. E, N[77,78]
		Tinaniformes	Tinanidae	Great tinamou, *Tinamous major* A,N[98]
		Trogoniformes	Trogonidae	Trogon, unid. sp.[78]
		Unspecified		Birds, unid. spp. E[78]
	Mammalia	Carnivora	Procyanidae*	Coatis, *Nasua narica*[75,76,77,78,81]
		Chiroptera		Bats, unid. spp.[75,77,78,81]
		Rodentia	Sciuridae*	Variagated squirrel, *Sciurus variagatoides*[75,76,77,78,81,82]
				Rodents, unid. spp.[48,50,51,52,78,77,78,81]
			Echimyidae	Spiny rats, *Phyllomys* sp.[75,76,77,78,81,82]
Cebus cay	Aves	Passeriformes	Furnariidae	Pale-legged hornero, *Furnarius leucopus* E,N[83]
		Pelicaniformes	Threskiornithidae	Plumbeous ibis, *Theristicus caerulescens* E,N[84]
	Mammalia	Rodentia	Cricetidae	Climbing mouse, *Rhipidomys* sp.[85]
Cebus olivaceous	Amphibia*	Anura		Frogs, frogs' eggs, unid. spp.[86]
	Reptilia	Squamata	Iguanidae	Iguanas, *Iguana iguana*[86]
	Aves*	Columbiformes	Columbidae	Ground dove, *Columbina talpacoti* E, A[86]
				White-tipped dove, *Leptotila verreauxi* E, N[86]
		Cuculiformes	Cuculidae	Greater ani, *Crotophaga major* E[86,87]
		Galliformes	Cracidae	Rufous-vented chacalaca, *Otralis ruficauda* E[86]
		Gruiformes	Rallidae	Grey-necked wood rail, *Aramides cajaneus* E[86]
		Opisthocomiformes	Opisthocomidae	Hoatzin, *Opisthocomus hoazin* E[86]
		Passeriformes	Tyrannidae	Golden-faced tyrannulet, *Zemmaria chrysops*[89]
		Pelicaniformes	Threskiornithidae	Green ibis, *Mesimbrinibis cayennensis* E[86]
		Struthioniformes	Tinamidae	Red-legged tinamou, *Crypturellus erythropus* E[86]
				Birds, unid. spp.[90]
	Mammalia	Rodentia	Sciuridae	Red-tailed squirrel, *Sciurus granatensis*[86]
Saimiri boliviensis	Reptilia*	Squamata		Lizards, unid. spp.[90]
	Amphibia*	Anura		Frogs, unid. spp.[90]
	Aves*			E,F,N unid. spp.[90]
Saimiri cassaquiarensis				"small vertebrates", spp. unid.[91]
Saimiri oerstedii	Amphibia	Anura*		Frogs, unid. spp.[91]
	Reptilia	Squamata*		Anoles, *Anolis* sp.[92]
	Aves*			E, N, unid. spp[91]
	Mammalia*	Chiroptera	Phyllostomidae	Tent-making bats, *Artibeus watsoni*[92]
Saimiri sciureus	Amphibia	Anura	Hylidae	Tree frogs, *Hyla* sp.[93]
				Lime tree frogs, *Sphaenorhynchus* sp.[93]
Sapajus apella	Amphibia	Anura*		Frogs, unid. spp.[63,95]
			Hyelidae	Tree frogs, $\geq$ 3 unid. species[101]
	Reptilia	Reptilia*	Squamata	Lizards, unid. spp.[63,95]
	Aves*	Columbiformes	Columbidae	Pale-vented pigeon, *Patagioenas cayennensis* E, N[96]
		Opisthocomiformes	Opithocomidae	Hoatzin, *Opithocomus hoazin* E,N[96]
		Passeriformes	Cardinalidae	Greyish saltator, *Saltator caerulescens* E,N[96]
			Grallariidae	Streak-chested antpitta, *Hylopezus perspicillatus* N[97]
			Icteridae	Yellow rumped cacique, *Cacicus cela* E, N[96]
				Troupial, *Icterus icterus* E, N[96]
			Thamnophilidae	Chestnut-backed antbird, *Myrmiceza excel*[98]
				Western slaty antshrike, *Thamnophilus atronucha* N[99]

(*continued on next page*)

Appendix 1 (*continued*)

Predator:	Prey: Class	Order	Family	Species
			Thraupidae	Red-capped cardinal, *Paroaria gularis*[96]
				Blue-gray tanager, *Thraupis episcopus* E.N[96]
			Trogloditidae	Banded wren, *Thyrothorus pleurostictus* N[96]
			Tyrannidae	Piratic flycatcher, *Legatus leucophaueus* E,N[96]
				Boat-billed flycatcher, *Megarhynchus pitangua* E,N[96]
				Social flycatcher, *Myiozetetes similis* E,N[96]
				Great kiskadee, *Pitangus sulphuratus* E,N[96]
				Tropical kingbird, *Tyrannus melancholicus* E,N[96]
		Pelicaniformes	Ardeidae	Heron, *Butorides* sp. E,N[100]
		Piciformes	Picidae	Helmeted woodpecker, *Dryocopus lineatus* E,N[96]
		Psittaciformes	Psittacidae	Chestnut-fronted macaw, *Ara severus* E,N[96]
				Dusky-headed parakeet, *Aratingella dellii* E,N[96]
		Tinnaniformes	Tinanidae	Great tinamou, *Tinamus major* N[98]
				Birds, unid. spp. A, E[63,101,102,103]
	Mammalia	Chiroptera	Emballanuridae	Proboscis bat, *Rhynchonycterus naso*[101]
		Didelphimorphia	Didelphidae	Opossum, *Didelphis* sp.[105]
		Primates	Aotidae	Owl monkeys, *Aotus lemurinus*[104]
			Pitheciidae	Brown titi monkey, *Callicebus brunneus*[106]
				Dusky titi monkeys, *Callicebus molloch*[106,107]
S. apella		Rodentia	Cricetidae	Arboreal rat, *Oecomys* sp.[95]
			Muridae	Mouse, *Mus* sp.[105]
				Rat, *Rattus rattus*[105]
			Sciuridae	Guianan squirrel, *Sciurus aestuans*[34,108]
				Northern Amazon red squirrel, *Sciurus igniventris*[95]
Sapajus flavius	Reptilia	Squamata	Iguanidae	Green iguana, *Iguana iguana*[109]
			Teiidae	Whiptail, *Kentropyx* sp.[109]
				Tegu, *Tupinambis* sp.[109]
Sapajus libidinosus	Reptilia	Squamata*	Colubridae	Vine snake, *Chironius* sp.; ≥2 other unid. spp.[110]
			Tropiduridae	*Tropidurus* spp.[111]
Sapajus nigritus	Aves	Pelicaniformes	Ardeidae	Night heron, *Nycticorax nycticorax* E,N[112]
		Piciformes	Picidae	Helmeted woodpecker, *Dryocopus lineatus* E,N[113]
Sapajus xanthosternos	Mammalia	Primates	Callitrichidae	Common marmosets, *Callithrix jacchus*[114,115]
Pitheciidae				
Cacajao calvus	Reptilia	Squamata		Small lizard, unid. sp.[116]
Cacajao melanocephala	Amphibia	Anura		Frog eggs, tadpoles, unid. spp.[117]
Cacajao ouakary	Reptilia	Chelonidae		Turtle eggs, unid. sp.[118]
Callicebus coimbrai	Aves	Passeriformes	Turdidae	Pale-breasted thrush, *Turus leucomelas* N[119]
Cercopithecoidea				
Cercopithecidae				
Allenopithecus nigriviridis	Teleostei			Fish, unid. spp.[120]
Cercocebus galeritus	Amphibia	Anura		Frogs, unid. spp.[121]
Cercopithecus ascanius	Aves	Columbiformes	Columbidae	African green pigeons, *Treron calva* A[122]
Cercopithecus campbelli	Amphibia	Anura		Tree frogs, unid. spp.[123]
Cercopithecus cephus	Aves			Birds, unid. spp. N[124]
Cercopithecus petaurista	Amphibia	Anura	Hylobatidae	Tree frogs, unid. spp.[123]
Cercopithecus diana	Amphibia	Anura	Hylobatidae	Tree frogs, unid. spp.[123]
Cercopithecus l'hoesti	Aves	Birds, unid. spp. E[125]		
Cercopithecus mitis	Reptilia	Squamata		Lizards, unid. spp.[126]
	Aves*	Galliformes	Numididae	Guinea fowl, sp. unid. E[127]
		Passeriformes	Cisticolidae	Gray-backed camaroptera, *Cameroptera brachyura*[128]
			Cisticolidae	Bar-throated apalis, *Apalis thoracica*[128]
			Locustellidae	Evergreen forest warbler, *Bradypterus barratti*[128]
			Pycnonotidae	Striped-cheek bulbul, *Andropadus milianjensis*[128]
		Strigiformes	Strigidae	African wood owl, *Strix woodfordi*[129]
				Birds, unid. spp.[130]
	Mammalia	Primates	Galagidae	Galago, *Galago senegalensis* or *Galagoides demidovii*[131]
		Rodentia	Muridae	Mice, unid. sp.[126]
			Sciuridae	Lord Derby's flying squirrel, *Anomalurus derbianus*[127]
				Squirrel, unid. sp.[132]
Cercopithecus neglectus	Teleostei			Fish, unid. spp.[120]
Chlorocebus pygerythrus	Aves*		Ploceidae	Masked weavers, *Plocinus* velatus F[133]
		Pelicaniformes	Ardeidae	Cattle egret, *Bulbulus ibis* E,F[133]
Chlorocebus sabaeus	Reptilia	Squamata	Agamidae	Common agama, *Agama agama*[259]
			Chamaeleonidae	Chameleons, unid. sp.[134]
	Aves	Bucerotiformes	Bucerotidae	Red-billed hornbill, *Tockus erythrorhynchus* E[135]

Appendix 1 (*continued*)

Predator:	Prey: Class	Order	Family	Species
		Columbiformes	Columbidae	Laughing dove, *Streptopelia senegalensis* E[135]
		Passeriformes	Passeridae	Northern grey-headed sparrow, *Passer griseus* E[135]
	Mammalia	Lagomorpha	Leporidae	Hares, *Lepus crawshayi*[135]
		Rodentia	Muridae	African grass rat, *Arvicanthis niloticus*[135]
Erthyrocebus patas	Reptilia	Squamata*	Agamidae	Agama lizards, *Agama agama*[136]
				Lizards, unid. spp.[137]
Macaca assamensis	Amphibia			Taxa unspecified[138]
	Reptilia			Taxa unspecified[138]
	Aves			Taxa unspecified[138]
	Mammalia			Taxa unspecified[138]
	Vertebrata			Taxa unspecified[139,140]
Macaca fascicularis	Teleostei			Fish, unid. sp.[141]
	Amphibia	Anura		Frogs, unid. spp.[142]
Macaca fuscata	Amphibia	Anura	Hylidae	Japanese tree frog, *Hyla japonica*[143]
			Ranidae	Tago's brown frog, *Rana tagoi*[143]
				Frogs, unid. spp.[144,145]
	Reptilia	Squamata	Scincidae	Far eastern skink, *Plestiodon latiscutatus*[143]
			Gekkonidae	Yakushima gecko, *Gekko yakuensis*[143]
		Squamata		Lizards, unid. spp.[144,145]
	Aves	Passeriformes	Zosteropidae	Warbling white-eye, *Zosterops japonica*[144]
Macaca leonina	Aves	Cuculiformes	Cuculidae	Coral-billed ground cuckoo, *Carpococcyx renauldi* N[146,147]
		Galliformes	Phasianidae	Siamese fireback, *Lophura diardi* E[147]
				Unid. spp., E, N[148]
	Mammalia	Rodentia	Sciuridae	Squirrels, unid. sp.[145]
Macaca nemestrina	Mammalia	Rodentia	Muridae*	Rats, *Rattus* sp.[149,150]
				"Vertebrates" (unspecified)[151]
Macaca nigra	Amphibia	Anura		Frogs, unid. spp.[152]
	Reptilia	Squamata	Gekkonidae	*Gekko* sp.[152]
	Aves	Galliformes	Phasianidae	Red jungle fowl, *Gallus gallus* E[152]
		Passeriformes	Monarchidae	Black-naped monarch flycatcher, *Hypothymes azurea* E[152]
	Mammalia	Chiroptera	Pteropodidae	Fruit bat, *Dobsonia* sp.[152]
Macaca radiata	Reptilia	Squamata		"Small lizards", unid. spp.[153]
Macaca silenus	Mammalia	Rodentia	Sciuridae	Indian giant quirrel, *Ratufa indica*[154]
Macaca sinica	Amphibia			Taxa unspecified[155]
	Reptilia			Taxa unspecified[155]
	Mammalia			Taxa unspecified[155]
Mandrillus leucophaeus	Amhibia	Anura		Frogs, sp. unid.[156]
	Mammalia	Rodentia	Muridae	small murid, sp. unid.[156]
Mandrillus sphinx	Amphibia	Anura		Frogs, unid. sp.[157]
	Reptilia	Testudines	Testudinidae	Tortoise, unid. sp.[157]
	Mammalia	Artiodactyla	Bovidae	Bay duiker, *Cephalophus dorsalis*[158]
		Eulipotyphla	Soricidae	White-toothed shrews, *Paracrocidura schoudeni*[158]
		Rodentia	Muridae	Swamp rat, *Malacomes longipes*[158]
Miopithecus talapoensis	Aves			Unid. spp A, E[124]
	Mammalia	Rodentia		Unid. spp.[124]
Papio anubis	Teleostei	Cypriniformes	Ciprinidae	Kuhe, *Boulengero microlepis*[158a]
				Fish, unid. sp.[158a]
	Reptilia	Squamata*		Snakes, unid. spp.[159]
			Gekkonidae	Gekkos, sp. unid.[160]
			Chamaeleonidae	Chameleons, unid. sp.[159,161]
			Scincidae	Skinks, unid. spp.[161]
			Varanidae	Monitors, *Varanus* sp.[161]
				Lizards, unid. spp.[161,162,163]
	Aves*	Charadriiformes	Turnicidae	Button quail, *Turnix sylvatica*[162]
		Galliformes	Numididae	Guinea fowl, unid. sp.[160]
			Phasianidae	Coqui francolin, *Peliperdix coqui* E, A[161]
				Birds, unid. spp. E, N, A[161,164,165]
	Mammalia*	Artiodactyla	Bovidae	Red duiker, *Cephalophus monticola*[166,167]
				Red-flanked duiker, *Cephalophus rufilatus*[168]
				Thomson's gazelle, *Eudorcas thomsoni*[159,161,162,164,165]
				Steenbok, *Raphiceros capestris*[162,164,165]
				Kirk's DikDik, *Madoqua kirki*[162,164,165]
				Bushbuck, *Tragelaphus scriptus*[160]
				Domestic goats, *Capra hircus*[169]
				Domestic sheep, *Ovis aries*[169]
		Chiroptera	Pterypodidae	Fruit bats, unid. sp.[160]
		Lagomorpha	Leporidae	Cape hare, *Lepus capensis*[159,161,162,164,165]
		Rodentia	Muridae	Rats, unid. spp.[160]
Papio cynocephalus	Amphibia	Anura		Frogs, unid. spp.[170]
	Reptilia	Squamata*		Lizards, unid. spp.[170]
				Snakes, unid. spp.[170]
	Aves*	Galliformes	Numididae	Helmeted guinea fowl, *Numida mitrata*[171]

(*continued on next page*)

Appendix 1 (*continued*)

Predator:	Prey: Class	Order	Family	Species
				Birds, unid. spp. E,N,A[170]
	Mammalia*	Artiodactyla	Bovidae	Impala, *Aepyceros melampus*[170,171,172]
				Grant's gazelle, *Gazella granti*[170,171,172]
				Thomson's gazelle, *Eudorcas thomsoni*[170,171,172]
				Bushbuck, *Tragelphus scriptus*[170,171,172]
				Wildebeeste, *Connochaetes taurinus*[170]
				Domestic goats, *Capra hircus*[173]
		Lagomorpha	Leporidae	Cape hare, *Lepus capensis*[170,171,172]
		Primates	Cercopithecidae	Vervets, *Chlorocebus pygerythrus*[170,171,172]
			Galagidae	Galagos, *Galago senegalensis*[170,171]
		Rodentia	Bathyergidae	mole rat, unid. sp.[171]
Papio hamadryas	Reptilia	Squamata	Lacertidae	Long-tailed lizard, *Lacastia longicaudata*[174]
	Aves	Galliformes	Phasianidae	Francolin, *Francolinus* sp.[174]
.				Helmeted guinea fowl, *Numida meleagris*[174]
				Birds, 3 unid. spp.[174]
	Mammalia	Artiodactyla	Bovidae	DikDik, *Madoqua saltiana*[174,175]
				Lesser kudu, *Strepsiceros imberis*[174]
		Hyracoidea	Procavidae	Rock hyrax, *Procavia habissinica*[174]
		Lagomorpha	Leporidae	Cape hare, *Lepus capensis*[174]
Papio papio	Aves			Birds, unid. spp.[176]
	Mammalia	Lagomorpha	Leporidae	Hare, *Lepus microtis*[176,259]
		Artiodactyla	Bovidae	Bushbuck, *Traglephus scriptus*[176]
Papio ursinus	Teleostei	Cypriniformes	Ciprinidae	Chubby head carp, *Barbus anoplus*[177]
				Common carp, *Cyprinus carpio*[177]
		Perciformes	Cichlidae	Tilapia, *Sarotherodon mossambicus*[177]
	Reptilia	Testudines	Testudinidae	Angulate tortoises, *Chersina angulata*[178]
	Aves*	Galliformes	Phasianidae	Francolin, *Francolinus* sp.[120]
				Birds, unid. spp. E, A[120]
	Mammalia*	Artiodacytla	Bovidae	Klipspringer, *Oreotragus oreotragus*[179,180,181,182]
				Domestic goats, *Capra hircus*[183]
				Domestic sheep, *Ovis aries*[183]
		Lagomorpha	Leporidae	Hare, *Lepus* capensis[179,180,181]
		Primates	Cercopithecidae	Green monkeys, *Chlorocebus sabaeus*[181]
		Rodentia	Sciuridae	Bush squirrel, *Paraxerus cepapi*[181]
Colobidae				
Rhinopithecus bieti	Aves			Birds, unid. spp. E[184]
	Mammalia	Rodentia	Anomaluridae	"Flying squirrel", unid. sp.[184]
Hominoidea				
Hylobatidae				
Hoolock leuconedys	Aves			Birds, unid. spp. N[185]
Hylobates lar	Aves			Birds, unid. spp.[186]
Nomascus concolor	Aves			Birds, 2 unid. species E, N[187,188]
Pongidae				
Pongo abelii	Mammalia	Primates	Lorisidae	slow lorises (*Nycticebus coucang*)[189,190]
Pongo pygmaeus	Teleostei	Siluriformes	Siluridae	Sheathed catfish, *Kryptopterus* spp., *Ompok* spp., *Wallago leeri*[191]
P. pygmaeus, cont.		Anabantiformes	Channidae	Snakeheads, *Channa* sp.[191]
	Mammalia	Rodentia	Sciuridae	horse-tailed squirrel, *Sundasciurus hippuras*[192]
	Mammalia	Rodentia	Muridae	"rat", unid. sp.[193]
Hominidae				
Pan paniscus	Aves			Birds, unid. sp.[194]
	Mammalia	Artiodactyla	Bovidae*	Bay duiker, *Cephalophus dorsalis*[194,195,196]
				Black-fronted duiker, *Cephalophus nigrifrons*[194,195]
				Blue duiker, *Philantomba monticola*[194,196]
				Weyn's duiker, *Cephalophus weynsi*[197]
				Duiker, unid. spp.[198,199]
			Suidae	Red river hog, *Potomocheoerus porcus*[200]
			Tragulidae	Water chevrotain, *Hyemoschus aquaticus*[200]
		Chiroptera		Bats, unid. spp.[199,201,202]
		Eulipotyphla		Shrew, unid. sp.[203]
		Primates	Cercopithecidae	Redtailed monkeys, *Cercopithecus ascanius*[194,204]
				Wolf's guenon, *C. wolfii*[194,204]
				Black mangabey, *Lophocebus atterimus*[194,204,205]
			Colobidae	Red colobus, *Piliocolobus* sp.[194,204]
			Galagidae	Galagos, *Galagoides demidovii*[194,204]
		Rodentia	Anomaluridae	*Anomalurus (?) sp.*[201,202]
			Sciuridae	Scaly tailed squirrel, *Eidolon* sp.[206]
				Forest giant squirrel, *Protoxerus stangeri*[200]
				Unidentified squirrels[199,202,206]
Pan troglodytes	Pisces			"small fish", sp. unid.[207]
	Reptilia	Testudines	Testudinidae	Tortoises, *Kinixys erosa*[208]
	Aves*	Accipitriformes	Accipitridae	Palm nut vulture, *Gypohierax angolensis* E,N[209]
				Black kite, *Milvus migrans* A[210]
		Buceratiformes	Buceratidae	Trumpeter hornbill, *Ceratogymna* buccinator N[210]
				Hornbill, unid. sp. A[211]

Appendix 1 (*continued*)

Predator:	Prey: Class	Order	Family	Species
		Cuculiformes	Cuculidae	White-browed coucal, *Centropus superciliosus*[212]
		Galliformes	Numididae	Crested guinea fowl, *Guttera eduardi*[212]
				Helmeted guinea fowl, *Guttera pucherani* A[209,213]
				Guinea fowl, unid. sp. A[160,214]
			Phasianidae	Domestic chicken, *Gallus gallus domesticus* A[212,213]
				Scaly francolin, *Francolinus squamatus* N[212]
				Francolin, *Francolinus* sp. N,A,J[160,209,210,214]
		Passeriformes	Colombidae	African green pigeon, *Treron australis* E[209]
			Corvidae	White-fronted crow, *Corvus alba* A[215]
				White-necked ravens, *Corvus alibicollis* N210
			Moticillidae	African pied wagtail, *Moticilla aguimp*[209]
			Ploceidae	Veillot's black weaver, *Ploceus nigerrimus* E,F[212]
				Weavers, ≥2 unid. spp.[209]
			Pycnonotidae	Yellow-vented bulbul, *Pycnotus barbatus*[209]
			Viduidae	Broad-tailed paradise whydah, *Steganura paradisea* A[216]
		Piciformes	Picidae	Woodpecker, unid. sp.[209]
		Strigiformes	Strigidae	African wood owl, *Strix woodfordi*[208]
		Unspecified		multiple species[160,214,215,216,217,218,219,220]
	Mammalia*	Artiodactyla*	Bovidae*	Red duiker, *Cephalophus natalensis*[212,213,222,223,224,225,226,227,228]
				Blue duiker, *Philantomba monticola*[213,214,220,222,226,227,228,229,230]
				Bushbuck, *Tragelaphus scriptus*[160,148,213,221,222,227,231,232]
				Klipspringer, *Oreotragus oreotragus*[233]
				Sitatunga, *Tragelaphus spekei*[234,209]
				Suni, *Neotragus moschatus*[233]
				Duiker, unid. sp.[210,235]
			Suidae*	Bushpigs*, *Potomochoerus larvatus*[160,209,212,213,214,217,218,219,220,223,225,226,227]
				Warthogs, *Phacochoerus aethiopicus*[212,223,225]
		Carnivora	Felidae	Leopard, *Panthera pardus*[236]
			Viverridae	African civet, *Viverra civetta*[210,212,223.225,]
				Banded mongoose, *Mungos mongo*[232]
				Mongoose, unid. sp.[212,223,225]
		Macroscelidea	Macroscelididae	Checkered elephant shrew, *Rhynchocyon cirnei*[212,223,225,228]
		Pholidota	Manidae	Pangolins, *Manis tricuspis*[208,237,238,236]
				Pangolin, unid. sp.[210]
	Primates *	Cercopithecidae*		[219,220,222,223,225,226,227,231,239]
				Campbell's guenon, *Cercopithecus campbelli*[235,241]
				Moustached guenon, *Cercopithecus cephus*[210]
				Diana monkey, *Cercopithecus diana*[221,242]
				L'Hoest's monkey, *Cercopithecus l'hoesti*[213,227,243]
				Blue monkey, *Cercopithecus mitis*[160,209,212,213,214,216,217,218,219,222,223,225,226,227,231,240,243,244,245]
				Mona monkey, *Cercopithecus mona*[221]
				Lesser spot-nosed guenon, *Cercopithecus petaurista*[221]
				Crowned guenon, *Cercopithecus pogonias*[210]
P. troglodytes, cont.				Sooty mangabey, *Cercocebus atys*[221,235,241]
				Vervet, *Chlorocebus pygerythrus*[160]
				Green monkey, *Chlorocebus sabaeus*[232,246]
				Patas monkey, *Erythrocebus patas*[232,247]
				Gray-cheeked mangabey, *Lophocebus albigena*[213,224,226,227]
				Olive baboon, *Papio anubis*[160,167,209,213,216,217,218,219,220,222,226,227,228,231]
				Yellow baboon, *Papio cynocephalus*[213]
			Colobidae*[a]	Black & white colobus, *Colobus angolensis*[245]
				Black and white colobus, *Colobus guereza*[213,222,226,227,228,229,244,243]
				Black and white colobus, *C. polykomos*[221,235,241]
				Black colobus, *Colobus satanas*[224]
				Red colobus, *Piliocolobus tephrosceles*[160,209,212,213,214,216,217,218,219,220,223,225,226,227,231,239,248,249,250,251,252,253,254,255,256]
				Red colobus, *Piliocolobus badius*[210,221,242,248,249]
				Olive colobus, *Procolobus verus*[221,242]
			Galagidae*	Senegal bushbaby, *Galago senegalensis*[212,214,222,223,225,257,258]
				Demidoff's bushbaby, *Galagoides demidovii*[227]
				Thick-tailed bushbaby, *Otolemur crassicaudatus*[212,214,223,225]
			Lorisidae	Pottos, *Perodicticus potto*[221,224,237,238,259]
		Rodentia*	Anomaluridae	Lord Derby's flying squirrel, *Anomalurus derbianus*[235,241]
				Flying squirrel, *Anomaluridae* unid. sp.[210]
			Muridae	Mice and rats, unid. spp.[106,237,238]
			Nesomyidae	Giant pouched rats, *Cricetomys eminii*[212,214,223,227]
			Sciuridae	Squirrels, *Funsciurus* sp.[225]
				Red-legged sun squirrel, *Heliosciurus rufobrachium*[261]
				Squirrels, *Protoxerus* sp.[212,214,223,225]
				Sun squirrels, *Heliosciurus* sp.[214,212,223,225]
				Squirrel, *H. rufobrachium* or *Protoxerus stangeri*[262]
			Thryonomidae*	Greater cane rats, *Thryonomys swinderianus*[212,214,223,227]

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[a] Some authors (e.g., Hosaka et al., 2020) refer to red colobus at Mahale as *Piliocolobus rufomitratus*.

Acknowledgments

I am immensely indebted to Chris Aligarya, Charles Birungi, Davis Kalunga, Diana Kamweri, Adolph Magoba, Godfrey Mbabazi, Braise Mugisha, Lawrence Ngandizi, Alfred Tumusiime, and Ambrose Twineomujuni for their invaluable assistance with collecting field data at Ngogo, some of which contributed to this paper. Fieldwork there would not have been possible without the logistical, administrative, and intellectual support of the late Jeremiah Lwanga and of Samuel Angedakin. I gratefully acknowledge the support and collaboration of John Mitani and Kevin Langergraber. I thank Makerere University, the Uganda Wildlife Authority, and the Uganda National Council for Science and Technology for permission to work in Kibale National Park. Data from Ngogo were collected with support from NSF grants BCS-0215622 and IOB-0516644, the L.S.B. Leakey Foundation, the National Geographic Society, and by Arizona State University, the Max Planck Institute for Evolutionary Anthropology, the University of Michigan, and Yale University. I thank 2 anonymous reviewers for their comments and constructive criticism on an earlier version of this manuscript, and especially thank Andrea Taylor for her constructive criticism and careful editing, and Bill McGrew for his comments, corrections, and encyclopedic command of the literature. I hope my response to the comments and criticisms has led to major improvements.

Supplementary Online Material

Supplementary online material to this article can be found online at https://doi.org/10.1016/j.jhevol.2020.102882.

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